

Comparable Small Area Estimates over Time

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Abstract

This document describes the calculation, using the available R software, of small area estimators based on area-level models that are stable along time and allow for statistical comparisons over different time instants. It starts describing the calculation of basic direct estimators and their sampling errors, for each time instant. These direct estimators and their sampling variances are the inputs for the production of EBLUPs (or EB) predictors under the area-level models. Hence, the latter estimators are reviewed when estimating at each time instant separately (i.e. cross-sectionally). Then, estimators that account for time correlation and others that account for spatial and temporal correlation are reviewed. For all the estimators, the steps of each of the estimation procedures are outlined, and these estimation procedures are illustrated with examples of application using open-access data. Finally, statistical comparison in two time instants is described.

Keywords: EBLUP, Empirical best, Poverty mapping, Small area estimation, Time correlation, Welfare estimation.

JEL Classification: C55, C87, C15.

1 Introduction

When estimation is desired for subpopulations (referred to domains or areas) of a population from which the sample is drawn, the sample sizes in some of those domains or areas might not allow for the production of estimates of acceptable quality. Small area estimation methods, making some kind of homogeneity assumptions across the areas, “borrow strength” from related areas to improve the efficiency of the disaggregated estimators. Perhaps the most widely applied small-area estimation procedure is the Fay-Herriot (FH) model ([Fay and Herriot, 1979](#)), which assumes an area-level linear regression model for the true indicators in the areas in a first stage. In a second stage, it models the sampling error of the survey-based direct estimators for these target quantities. In the simplest case of no transformations taken on the target quantities, the two-stage model leads to a single area-level linear mixed model that incorporates random area effects, with known heteroscedastic error variances. This procedure only requires data at the area-level, often avoiding legal data protection issues.

Consider that small area estimates of living conditions are desired for several time instants. The existing inertia in the socio-economic conditions of each target area due to its own history or idiosyncrasy, makes the true values of the indicator for the same area at different times related. However, estimators obtained from separate models applied for each time instant (cross-sectionally) do not incorporate this correlation and do not make use of the historical information. Therefore, they do not “borrow strength” from time, providing unstable estimates along time that are not comparable, in the sense of not allowing to statistically distinguish the changes in the target indicators in two time points.

This document reviews small area estimation procedures that provide, using the available R software, comparable and stable estimates over time, based on extensions of the basic FH model to the multivariate case. The resulting estimators account for time correlation and hence “borrow strength” from time, as well as from the other areas. These procedures build on the basic estimators and basic small area models. Hence, after introducing the notation and the target indicators (Section 2), these basic procedures are first reviewed for the estimation at a fixed time point (Sections 3 and 4). Later, the multivariate extension of the FH model proposed by Benabent and Morales (2016) is introduced in Section 5. This model includes several specifications for the involved covariance matrices, two of them accounting for time-correlated area effects. One of these specifications considers homoscedastic area effects over time, and another allows for heteroscedastic area effects over time. Finally, a spatio-temporal model is also described in Section 6. Each estimation procedure is described and illustrated with examples of application, providing the R codes that users can reproduce themselves. Section 7 comments on how to perform statistical comparison for two time points. Finally, some concluding remarks are provided in Section 8.

2 Notation and target indicators at a fixed time

Consider a finite population U of size N at a given time period t . This population is supposed to be partitioned into D subpopulations, called areas or domains, $U_1, \dots, U_D$, with population sizes $N_1, \dots, N_D$. Let y_{di} be the value of the target variable for the i -th individual within the d -th area, for $d = 1, \dots, D$ and $i = 1, \dots, N_d$. In this paper, we wish to estimate additive area indicators defined as

$$\delta_d = \frac{1}{N_d} \sum_{i=1}^{N_d} h(y_{di}), \quad d = 1, \dots, D, \quad (1)$$

for a general real-valued function $h(\cdot)$. Simple indicators are obtained when $h(\cdot)$ is a linear function. For the identity function $h(y) = y$, the target area indicators are the area means of the y_{di} -values, $\delta_d = \bar{Y}_d = N_d^{-1} \sum_{i=1}^{N_d} y_{di}$, $d = 1, \dots, D$.

Additive indicators of great interest in this document are the FGT poverty indicators (Foster et al., 1984). Let E_{di} be a welfare measure (income, expenditure) for individual i in area d , and let z be the poverty line, defined for EU countries as $z = 0.6 \text{ Median}(E_{di})$, where the median is taken across all the areas of the population in that time instant t . The FGT family of poverty indicators for area d is defined as

$$F_{\alpha d} = \frac{1}{N_d} \sum_{i=1}^{N_d} I\left(\frac{z - E_{di}}{z}\right)^\alpha I(E_{di} < z), \quad d = 1, \dots, D, \quad \alpha \geq 0, \quad (2)$$

where $I(E_{di} < z) = 1$ if $E_{di} < z$ and $I(E_{di} < z) = 0$ otherwise. Denoting

$$F_{\alpha di} = I\left(\frac{z - E_{di}}{z}\right)^\alpha I(E_{di} < z), \quad i = 1, \dots, N_d,$$

the FGT poverty indicator is the area mean of the $F_{\alpha di}$ -values, that is,

$$F_{\alpha d} = N_d^{-1} \sum_{i=1}^{N_d} F_{\alpha di}.$$

The FGT indicator obtained for $\alpha = 0$, F_{0d} , is the poverty rate in area d , which is an indicator of special interest.

For a given time period t , inference on the target indicators δ_d , $d = 1, \dots, D$, is based on a sample s_d of size $n_d < N_d$ drawn independently from each area U_d at that time period. Then, $s = s_1 \cup \dots \cup s_D$ is the overall sample, of size $n = \sum_{d=1}^D n_d < N$. The next section describes direct estimators for the areas at a fixed time point t .

3 Direct estimators at a fixed time

In this section, we omit the subscript t in the notation, although everything that follows is referred to a given time instant t . We introduce the basic direct estimators of the area means $\delta_d = \bar{Y}_d$ at time t , which use only the survey data from the area of interest referred to that time instant. Direct estimators are typically based on survey sampling theory. In that theory, the values y_{di} are assumed to be fixed (not random), and hence no model assumptions are made for y_{di} . Hence, randomness arises only from the mechanism used to draw the sample (the sampling design). Direct estimators exhibit good design properties for areas with sufficiently large sample sizes, but they are inefficient, in the sense of having large sampling variances, for areas with small sample sizes. Corral et al. (2022), Section 2.1, introduces the design-based inference setup in the context of SAE for poverty mapping. For a much more detailed account of the sampling theory, see Cochran (2022), Särndal et al. (1992), Thompson (1997), or Lohr (2010).

Sampling theory was traditionally developed for linear indicators. Here we describe traditional direct estimators for area means

$$\bar{Y}_d = N_d^{-1} \sum_{i=1}^{N_d} y_{di}, \quad d = 1, \dots, D.$$

An unbiased estimator of $\bar{Y}_d$ across all the possible samples s_d drawn from U_d using the specified sampling design is the Horvitz-Thompson (HT) estimator. Let π_{di} be the inclusion probability of unit i in the sample s_d from area d , and $w_{di} = \pi_{di}^{-1}$ be the survey weight (or elevation factor) of that unit. The survey weight is interpreted as the number of population units from area d that the i -th sample unit from s_d represents. The Horvitz-Thompson estimator of $\bar{Y}_d$ is then given by

$$\hat{Y}_d = N_d^{-1} \sum_{i \in s_d} w_{di} y_{di}. \quad (3)$$

If $\pi_{di} > 0$ for all $i = 1, \dots, N_d$, the variance across the possible samples s_d (hereafter design variance) of the HT estimator of $\bar{Y}_d$ is given by

$$\text{var}_{\pi}(\hat{Y}_d) = \frac{1}{N_d^2} \left\{ \sum_{i=1}^{N_d} \frac{y_{di}^2}{\pi_{di}} (1 - \pi_{di}) + 2 \sum_{i=1}^{N_d} \sum_{j>i}^{N_d} \frac{y_{di} y_{dj}}{\pi_{di} \pi_{dj}} (\pi_{d,ij} - \pi_{di} \pi_{dj}) \right\}, \quad (4)$$

where $\pi_{d,ij}$ is known as the second-order inclusion probability of units $i \in U_d$ and $j \in U_d$, which is the joint inclusion probability of those units in the sample s_d , for $i \neq j$. Note that $\pi_{d,ii} = \pi_{di}$, for all $i = 1, \dots, N_d$.

If additionally $\pi_{d,ij} > 0$ for all $i \in U_d$ and $j \in U_d$, an unbiased estimator of the sampling variance of the HT estimator $\hat{Y}_d$ of $\bar{Y}_d$ is given by

$$\widehat{\text{var}}_{\pi}(\hat{Y}_d) = N_d^{-2} \left\{ \sum_{i \in s_d} \frac{y_{di}^2}{\pi_{di}^2} (1 - \pi_{di}) + 2 \sum_{i \in s_d} \sum_{j>i \in s_d} \frac{y_{di} y_{dj}}{\pi_{di} \pi_{dj}} \left(\frac{\pi_{d,ij} - \pi_{di} \pi_{dj}}{\pi_{d,ij}} \right) \right\}. \quad (5)$$

Sometimes, in the estimation phase, no other design information apart from the survey weights w_{di} is available to the statistician. In this case, the unbiased estimator (4) cannot be calculated. A simple approximation that avoids the use of second-order inclusion probabilities is obtained by considering $\pi_{d,ij} \approx \pi_{di} \pi_{dj}$, for $i \neq j$. Under Poisson sampling, the equality $\pi_{d,ij} = \pi_{di} \pi_{dj}$, for $i \neq j$, holds and the second term on the right-hand side in (5) becomes zero. Replacing then $w_{di} = \pi_{di}^{-1}$ in the resulting

expression, we obtain the following estimator of the sampling variance of the direct HT estimator $\hat{Y}_d$, which does not depend on the second-order inclusion probabilities,

$$\widehat{\text{var}}_{\pi}(\hat{Y}_d) = N_d^{-2} \sum_{i \in s_d} w_{di}(w_{di} - 1)y_{di}^2. \quad (6)$$

This estimator is the one returned by the function `direct()` of the R package `sae`, which will be used in Example 1 of application of the above procedures, when survey weights are considered. This function assumes that there is no information about the sampling design apart from the survey weights. In the case of having additional information about the sampling design, we recommend user the much more appropriate R packages `survey` (Lumley, 2004) or `sampling` (Tillé and Matei, 2012). There are also alternative approximations of the sampling variance, depending on the sampling design and the available information, see e.g. the method of the balanced repeated replications (BRR) with the Fay correction (US Bureau of Labor Statistics and US Census Bureau, 2006).

Although the HT direct estimator $\hat{Y}_d$ is unbiased for $\bar{Y}_d$, it is not a weighted average unless $N_d = \sum_{i \in s_d} w_{di}$, and it may have larger variance than the ratio HT (or Hájek) estimator of $\bar{Y}_d$, defined as the weighted average

$$\hat{Y}_d^R = \frac{1}{\hat{N}_d} \sum_{i \in s_d} w_{di}y_{di} = \bar{y}_{dw}, \quad (7)$$

where $\hat{N}_d = \sum_{i \in s_d} w_{di}$. The design variance of the Hájek estimator $\hat{Y}_d^R$ may be obtained by applying the Taylor linearization method to the ratio of HT estimators $\hat{N}_d$ and $\hat{Y}_d^R = \sum_{i \in s_d} w_{di}y_{di}$ of the totals N_d and $Y_d = \sum_{i=1}^{N_d} y_{di}$, respectively.

In practice, weights w_{di} provided in surveys are calibrated so that the HT estimates of certain available totals agree with the true totals, which leads to calibration estimators (Deville and Särndal, 1992), a special case of which is the well-known generalized regression (GREG) estimator. Calibrated weights are typically further adjusted for non-response, and the original survey weights are seldom provided in the survey data. Then, adjusted direct estimators obtained by expansion as in (7) are typically used. Estimation of their sampling errors is more complex and falls out of the scope of this document, but the reader is referred to Fuller (1975) or Estevao et al. (1995).

If the sample s_d is drawn by simple random sampling (SRS), then $w_{di} = N_d/n_d$ for all $i = 1, \dots, N_d$, and both the HT direct estimator and the ratio HT estimator reduce to the usual (unweighted) sample mean,

$$\hat{Y}_d = \hat{Y}_d^R = n_d^{-1} \sum_{i \in s_d} y_{di} = \bar{y}_{ds}.$$

Under SRS without replacement, the design variance (4) reduces to the usual formula,

$$\text{var}_{\pi}(\hat{Y}_d) = (1 - f_d) \frac{\sigma_y^2}{n_d}, \quad \sigma_y^2 = \frac{1}{N_d - 1} \sum_{i=1}^{N_d} (y_{di} - \bar{Y}_d)^2, \quad (8)$$

where $f_d = n_d/N_d$ is the sampling fraction in area d , which equals the (constant) inclusion probabilities for the units under this sampling design.

In the case of binary target variable $y_{di} \in \{0, 1\}$, the HT estimator of the population proportion $P_d = N_d^{-1} \sum_{i=1}^{N_d} y_{di}$, reduces to the (unweighted) sample proportion

$$\hat{P}_d = n_d^{-1} \sum_{i \in s_d} y_{di} \doteq p_{ds}.$$

Moreover, the sampling variance of the sample proportion $\hat{P}_d = p_{ds}$ in (4) reduces to

$$\text{var}_\pi(\hat{P}_d) = (1 - f_d) \frac{P_d(1 - P_d)}{n_d} \frac{N_d}{N_d - 1} \approx (1 - f_d) \frac{P_d(1 - P_d)}{n_d},$$

for N_d large, which reduces to $P_d(1 - P_d)/n_d$ for negligible sampling fraction f_d .

For the FGT poverty indicator $F_{\alpha d}$, the HT estimator reads

$$\hat{F}_{\alpha d} = N_d^{-1} \sum_{i \in s_d} w_{di} F_{\alpha, di}. \quad (9)$$

and the variance estimator given in (6) becomes

$$\widehat{\text{var}}_\pi(\hat{F}_{0d}) = N_d^{-2} \sum_{i \in s_d} w_{di}(w_{di} - 1) F_{0, di}^2. \quad (10)$$

In practice, one needs to decide which direct estimator to use, but the most important choice can be whether to consider the survey weights. Survey-weighted direct estimators are consistent as n_d tends to infinity for general sampling designs. Hence, they protect from informative sampling, that is, when the selections are related with the target variable. Note that if richer (or poorer) individuals are selected to the sample with larger probabilities, then they should be overweighted to obtain unbiased estimators. However, they also inflate the variance of the estimators. Now if survey-weighted and unweighted estimators are very similar for all the areas, this means that survey weights have small contribution. This can be checked by plotting the survey-weighted direct estimates (HT or Hájek) against unweighted direct estimates (sample means), and the line $y = x$. If points stick to the $y = x$ line, survey weights may be ignored.

Direct estimation procedure:

Step 1 Decide which direct estimator to use. By default, survey weights are used to ensure good design properties.

Step 2 Compute the selected direct area estimators and estimators of their corresponding sampling variances.

The function `direct()` of the R package `sae` (Molina and Marhuenda, 2009) derives the direct HT estimators (3) and their estimated sampling errors approximated as (6). The call to this function is

```
direct(y, dom, sweight, domsize, data, replace = FALSE)
```

where we need to specify `y`, the vector with the individual values of the variable for which we want to estimate the domain means, `dom`, the vector or factor (of the same size as `y`) with the domain codes `sweight`, the optional vector with survey weights, and `domsize`, a $D \times 2$ data frame with the domain codes in the first column, and the corresponding domain population sizes in the second column. The argument `data` is an optional data frame containing the variables specified in `y`, `dom` and `sweight`. Finally, `replace` is by default specified to `FALSE` for random sampling without replacement within each domain, and should be changed to `TRUE` in the case of random sampling with replacement.

The function returns a $D \times 5$ data frame including `Domain`, the domain codes in ascending order, `SampleSize`, the domain sample sizes, `Direct`, the direct estimators of the domain means of the variable `y`, `SD`, the estimated standard deviations of the direct estimators, and `CV`, the estimated CVs, obtained as 100 times the standard deviation, by the absolute value of the corresponding direct estimate.

Example 1 below illustrates the calculation of direct HT estimators of area poverty rates at a given time t with a synthetic data set included in the `sae` R package (Molina and Marhuenda, 2009), applying the function `direct()`.

Example 1 (Direct HT estimators of poverty rates in R package sae) We illustrate the calculation of direct Horvitz-Thompson (HT) estimators of poverty rates using a synthetic data set on living conditions in Spanish provinces, included in the R package `sae`, called `incomedata`. The data contains information for $n = 17,119$ fictitious individuals residing across $D = 52$ Spanish provinces. The variables include the name of the province of residence (`provlab`), province code (`prov`), autonomous community code (`ac`), age group (`age`), taking values from 1 to 5, having Spanish nationality (`nat`), where 1 is for Spanish and 2 non-Spanish, education level (`educ`), ranging from 0 for individuals under 16 to 2 for university level, and employment status (`labor`), where 0 is for individuals below 16, 1 for employed, 2 for unemployed, and 3 for inactive. It also contains variables to distinguish whether the individual is in any of the age groups, from group 2 to group 5 (`age2` to `age5`), whether they have education level 1 to 3 (`educ1` to `educ3`), whether they hold Spanish nationality (`nat1`), whether they are employed, unemployed, or inactive (`labor1` to `labor3`), the equivalent net income (`income`), and the sampling weight (`weight`).

After installing the `sae` library, we load it together with the `incomedata` dataset, which contains the sample data, and the `sizeprov` dataset, which contains the population sizes of the $D = 52$ Spanish provinces, N_d :

```
library(sae)
data(incomedata)
attach(incomedata)
data(sizeprov)
```

Step 1: We will use the direct HT estimators that use the survey weights in `weight`. First, we calculate the total sample size, the number of provinces, the sample sizes for each province, and extract the population sizes from the `sizeprov` file:

```
n<-dim(incomedata)[1]      # Total sample size
D<-length(unique(prov))    # Number of provinces (areas or domains)
nd<-as.vector(table(prov)) # Sample sizes of the provinces
Nd<-sizeprov$Nd            # Population sizes of the provinces
```

We set the poverty line, calculated previously as $0.6 \times \text{median}(\text{income})$, and construct the variable `poor`, the indicator of having income below the poverty line:

```
z<-6557.143
poor<-numeric(n)
poor[income<z]<-1
```

Step 2: We now calculate the direct HT estimators of poverty rates in the provinces (averages of the variable `poor` in the provinces), using the `direct()` function, using the survey weights in `weight`:

```
povinc.dir.res<-direct(y=poor,dom=prov,sweight=weight,domsize=sizeprov[, -1])
print(povinc.dir.res,row.names=F)
```

The output of this function is:

Domain	SampSize	Direct	SD	CV
1	96	0.25503732	0.04846645	19.003670
2	173	0.14059242	0.03042195	21.638397
3	539	0.20785096	0.02178689	10.481979
4	198	0.26763976	0.04090335	15.282986
5	58	0.05512200	0.02555426	46.359465

6	494	0.21553890	0.02357906	10.939585
7	634	0.09999792	0.01536517	15.365488
8	1420	0.29812535	0.01618508	5.428952
9	168	0.21413150	0.04473542	20.891562
10	282	0.27031324	0.03125819	11.563692
11	398	0.14887351	0.02189022	14.703904
12	118	0.17598199	0.03584882	20.370731
13	250	0.20921534	0.03279230	15.673948
14	224	0.29975708	0.03934080	13.124228
15	495	0.25347550	0.02467716	9.735520
16	92	0.26334059	0.05913385	22.455274
17	142	0.18337421	0.03710194	20.232911
18	208	0.31727340	0.04043964	12.745990
19	89	0.17908182	0.04234025	23.642966
20	285	0.23690549	0.03194779	13.485457
21	122	0.12583449	0.03202547	25.450474
22	115	0.24107606	0.04856351	20.144476
23	232	0.31294198	0.04122671	13.173916
24	218	0.18801572	0.03002634	15.970122
25	130	0.15559590	0.03872448	24.887854
26	510	0.25811811	0.02459196	9.527405
27	173	0.37718722	0.05696330	15.102129
28	944	0.18218209	0.01639018	8.996593
29	379	0.22918462	0.02735631	11.936364
30	885	0.17703167	0.01648910	9.314210
31	564	0.16190765	0.01842017	11.376958
32	129	0.22799612	0.04199465	18.419018
33	803	0.26064010	0.02093779	8.033220
34	72	0.30166074	0.07179782	23.800849
35	472	0.16651843	0.02307258	13.855869
36	448	0.18549072	0.02418887	13.040474
37	164	0.16104513	0.02998243	18.617410
38	381	0.18429619	0.02054550	11.148085
39	434	0.34244429	0.03248937	9.487491
40	58	0.22262002	0.05639965	25.334492
41	482	0.20503036	0.02122527	10.352256
42	20	0.02541207	0.02540651	99.978151
43	134	0.32035438	0.04934077	15.401934
44	72	0.27364239	0.06723440	24.570172
45	275	0.12553377	0.02131991	16.983409
46	714	0.21360678	0.02070508	9.693081
47	299	0.19292332	0.03211484	16.646429
48	524	0.21694466	0.02215645	10.212948
49	104	0.30027442	0.06025302	20.065986
50	564	0.10034577	0.01569138	15.637311
51	235	0.19724796	0.03341193	16.939048
52	180	0.19109119	0.03441016	18.007191

Finally, we store the estimated values in a vector and count how many provinces have a coefficient of variation (CV) above 20%:

```
povinc.dir.cv<-povinc.dir.res$CV
sum(povinc.dir.cv>20)
```

We obtain that 15 provinces have direct HT estimators of poverty rate with estimated CV exceeding 20%. We will see that model-based procedures provide more efficient estimators for most of the provinces.

4 Fay-Herriot estimators at a fixed time

This section describes the Fay-Herriot (FH) model introduced by [Fay and Herriot \(1979\)](#) to “borrow strength” from other small areas and obtaining more efficient estimators than the direct ones. As in the previous section, we describe the procedures for a given time t (cross-sectionally), and the process can then be identically repeated for $t = 1, \dots, T$. Nevertheless, even though what follows is referred to a given time instant t , for notational simplicity, we omit the subscript t everywhere.

The FH model utilizes only aggregated data at the area level, which makes it broadly applicable. Probably for that reason, together with its simplicity, interpretability, and the favorable properties of the resulting estimators, it is possibly the most popular SAE model. For relevant applications and uses of this model in official statistics, specially in estimation of welfare indicators, we refer to [Molina et al. \(2009\)](#).

The model is defined in two stages. In the first stage, [Fay and Herriot \(1979\)](#) assume that the true area indicators δ_d vary linearly with a $p \times 1$ vector $\mathbf{x}_d = (x_{d1}, x_{d2}, \dots, x_{dp})'$ of area-level covariates, as follows:

$$\delta_d = \mathbf{x}_d' \boldsymbol{\beta} + u_d, \quad d = 1, \dots, D. \quad (11)$$

This model is known as *linking model*, because the vector of regression coefficients, $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_{p-1})'$, is assumed to be constant for all the areas, thus “linking” the areas. The simplest FH model assumes $u_d \stackrel{iid}{\sim} N(0, \sigma_u^2)$, with $\sigma_u^2 > 0$ unknown, and the terms u_d are referred to as *area effects*, because they model the between-area heterogeneity that is not explained by the area-level covariates in $\mathbf{x}_d$. Typically, $x_{d1} = 1$ to allow for an intercept in the regression and $x_{d2}, \dots, x_{dp}$ are the population means of $p - 1$ auxiliary variables, obtained from census data. In the case of a binary auxiliary variable indicating the presence of a characteristic, the population mean is actually the population proportion of this characteristic.

Note that the true indicators δ_d are not observable and hence model (11) cannot be fit. However, direct estimators $\hat{\delta}_d$, $d = 1, \dots, D$, are assumed to be available from the unit-level data in a (current) survey, where the areas may be identified. [Fay and Herriot \(1979\)](#) proposed to account for the sampling errors of the direct estimators $\hat{\delta}_d$ of δ_d , by considering the following model, known as *sampling model*, in the second stage:

$$\hat{\delta}_d = \delta_d + e_d, \quad d = 1, \dots, D. \quad (12)$$

In the simplest FH model, the sampling errors e_d , $d = 1, \dots, D$, are supposed to be independent of the area effects u_d , following $e_d \stackrel{ind}{\sim} N(0, \sigma_{ed}^2)$, where the sampling variances $\sigma_{ed}^2 = \text{var}_\pi(\hat{\delta}_d | \delta_d)$, $d = 1, \dots, D$, are assumed to be known and positive.

Replacing the linking model (11) in the sampling model (12) results in a single linear mixed model

$$\hat{\delta}_d = \mathbf{x}_d' \boldsymbol{\beta} + u_d + e_d, \quad d = 1, \dots, D, \quad (13)$$

with known and heteroscedastic error variances $\sigma_{ed}^2 > 0$, $d = 1, \dots, D$. For σ_u^2 known, [Henderson \(1950\)](#) obtained the best linear unbiased predictor (BLUP) of a linear combination of the vector of fixed effects $\boldsymbol{\beta}$ and the vector of random effects $\mathbf{u} = (u_1, \dots, u_D)'$ in a linear mixed model. The BLUP of $\delta_d = \mathbf{x}_d' \boldsymbol{\beta} + u_d$ is the predictor $\tilde{\delta}_d$ that is a linear function of $\hat{\boldsymbol{\delta}} = (\hat{\delta}_1, \dots, \hat{\delta}_D)'$, which is unbiased in the sense $E(\tilde{\delta}_d - \delta_d) = 0$, and minimizes the model MSE, defined as $E(\tilde{\delta}_d - \delta_d)^2$. Here, the expectations are taken

with respect to the distribution induced by the FH model (13), where the normality assumptions are not really necessary. However, now the true values δ_d are assumed to be random variables, unlike in the design-based inference. That is the reason why the term “estimator” of δ_d is replaced by “predictor” of (the realized value of) δ_d .

Based on the FH model (11)–(12), the BLUP of $\delta_d = \mathbf{x}_d' \boldsymbol{\beta} + u_d$ is obtained simply by fitting the linear mixed model (13) assuming σ_u^2 known, and then using the predicted value of δ_d through the model, that is,

$$\tilde{\delta}_d^{FH} = \mathbf{x}_d' \tilde{\boldsymbol{\beta}} + \tilde{u}_d. \quad (14)$$

Here, $\tilde{u}_d = \gamma_d(\hat{\delta}_d - \mathbf{x}_d' \tilde{\boldsymbol{\beta}})$ is also the BLUP of the area effect u_d , where $\gamma_d = \sigma_u^2 / (\sigma_u^2 + \sigma_{ed}^2)$, and $\tilde{\boldsymbol{\beta}}$ is the weighted least squares (WLS) estimator of $\boldsymbol{\beta}$, which equals the maximum likelihood (ML) estimator of $\boldsymbol{\beta}$ under normality of random effects and errors, given by

$$\tilde{\boldsymbol{\beta}} = \left(\sum_{d=1}^D \gamma_d \mathbf{x}_d \mathbf{x}_d' \right)^{-1} \sum_{d=1}^D \gamma_d \mathbf{x}_d \hat{\delta}_d. \quad (15)$$

Replacing $\tilde{u}_d = \gamma_d(\hat{\delta}_d - \mathbf{x}_d' \tilde{\boldsymbol{\beta}})$ in (14) and re-arranging the terms, the BLUP can be seen as a composite estimator between the direct estimator $\hat{\delta}_d$ used as response variable, and the above regression-synthetic estimator, that is,

$$\tilde{\delta}_d^{FH} = \gamma_d \hat{\delta}_d + (1 - \gamma_d) \mathbf{x}_d' \tilde{\boldsymbol{\beta}}, \quad (16)$$

where the weight attached to the direct estimator, $\gamma_d = \sigma_u^2 / (\sigma_u^2 + \sigma_{ed}^2) \in (0, 1)$, depends on the estimated variance σ_{ed}^2 of the direct estimator (which grows as the area sample size n_d decreases), relative to the unexplained area heterogeneity measured by σ_u^2 . For an area d where the direct estimator $\hat{\delta}_d^{DIR}$ is efficient, $\tilde{\delta}_d^{FH}$ attaches more weight to the direct estimator $\hat{\delta}_d$ than to the regression-synthetic counterpart, and more weight is given to the latter for the areas whose direct estimator has bad quality in terms of σ_{ed}^2 , relative to the unexplained between-area variability σ_u^2 . Note that the regression-synthetic estimator “borrows strength” from all the areas through $\tilde{\boldsymbol{\beta}}$. Hence, the BLUP (16) based on the FH model (13) automatically “borrows strength” for the areas where it is needed, but gets close to the direct estimator when it is efficient enough, which is a desirable property, given that direct estimators are design unbiased (or design consistent) without making any model assumptions. As a consequence, the BLUP (16) inherits the good (design) properties of the direct estimator for sufficiently large n_d , such as design consistency as n_d grows. The expression (16) indicates that the BLUP may be employed for all the areas $d = 1, \dots, D$, regardless of whether they have large or small sample size n_d ; or equivalently, whether their corresponding direct estimator has acceptable quality or not.

The BLUP of δ_d depends on the area effects variance, σ_u^2 , which is unknown in practice, and must be estimated. The usual estimation methods are ML and restricted/residual ML (REML), both using the Normal likelihood, or the FH method proposed by [Fay and Herriot \(1979\)](#), which is a moments method. The REML method corrects the ML estimator by accounting for the degrees of freedom associated with estimating the vector of regression coefficients $\boldsymbol{\beta}$, resulting in an estimator of σ_u^2 with smaller bias for finite number of areas D . Under certain regularity assumptions, the three estimators are consistent as the number of areas D grows.

Let $\hat{\sigma}_u^2$ be a consistent estimator of σ_u^2 . By replacing σ_u^2 with $\hat{\sigma}_u^2$ in the BLUP given in (14), we obtain the empirical BLUP (EBLUP) of δ_d ,

$$\hat{\delta}_d^{FH} = \hat{\gamma}_d \hat{\delta}_d + (1 - \hat{\gamma}_d) \mathbf{x}_d' \hat{\boldsymbol{\beta}}, \quad (17)$$

where now $\hat{\gamma}_d = \hat{\sigma}_u^2 / (\hat{\sigma}_u^2 + \sigma_{ed}^2)$ and $\hat{\boldsymbol{\beta}} = \left(\sum_{d=1}^D \hat{\gamma}_d \mathbf{x}_d \mathbf{x}_d' \right)^{-1} \sum_{d=1}^D \hat{\gamma}_d \mathbf{x}_d \hat{\delta}_d$. We will refer to the EBLUP (17) based on the FH model (13) as the FH estimator.

Under the normality of random effects u_d and sampling errors e_d and certain regularity assumptions, [Prasad and Rao \(1990\)](#) obtained an approximation with $o(D^{-1})$ error for large number of areas D for the MSE of the FH estimator, see [Rao and Molina \(2015\)](#), eqn. (6.2.1). They further obtained an estimator of the MSE with bias of error $o(D^{-1})$, see [Rao and Molina \(2015\)](#), pp. 136-137.

The FH model requires, apart from the direct estimators, which act as response variables, their sampling variances $\sigma_{ed}^2 = \text{var}_\pi(\hat{\delta}_d|\delta_d)$, $d = 1, \dots, D$. In practice, these variances are unknown and must be estimated based on the area-specific survey data. Since these direct estimates $\widehat{\text{var}}_\pi(\hat{\delta}_d|\delta_d)$ are unstable for small areas, it is customary to apply a smoothing procedure to those estimates. This can be done by the Generalized variance function (GVF) proposed by [Vaillant \(1987\)](#), or by a more general preliminary regression model for the direct estimates of the variances $\widehat{\text{var}}_\pi(\hat{\delta}_d|\delta_d)$. The smoothed variances obtained as predicted values from this model are then treated as the true variances σ_{ed}^2 .

We might encounter zero estimate $\widehat{\text{var}}_\pi(\hat{\delta}_d|\delta_d)$ for certain areas very small area sample sizes n_d . For these areas, a predicted value through the mentioned preliminary regression model for the direct estimators of the variances may be used as σ_{ed}^2 .

Consider now the case of a non-sampled area (with $n_d = 0$). In this case, γ_d , and hence the EBLUP, are not defined. In these areas, it is possible to use the regression part of the BLUP in (14) to estimate δ_d , which is known as the regression-synthetic estimator, $\tilde{\delta}_d^{RSYN} = \mathbf{x}_d' \hat{\beta}$. However, since the model cannot be checked for these areas and we can never be sure that these areas are not outlying, the general recommendation is not to produce estimates.

Below we list the steps of the procedure for the calculation of FH estimators and their estimated MSEs.

FH estimation procedure:

- Step 1** Compute the selected direct area estimators $\hat{\delta}_d$, $d = 1, \dots, D$, and estimators of their corresponding sampling variances (perhaps previously smoothed), denoted as σ_{ed}^2 , $d = 1, \dots, D$.
- Step 2** Select the aggregated auxiliary variables at the area level for the FH model. A simple approach is to perform a model selection procedure in a linear regression model without the area effects, e.g., using stepwise selection, exhaustive search or LASSO, using preferably BIC goodness-of-fit measure to compare nested models.
- Step 3** Fit the model and check the FH model assumptions, including linearity, normality of predicted area effects and standardized model residuals, and the presence of outlying areas.
- Step 4** In case of clear systematic model departures, the model should be changed. In case of isolated departures because of outlying areas, either do not obtain the FH estimate for those areas or change some aspect of the model or the data. Then, repeat Steps 2 and 3.
- Step 5** If the model assumptions hold, using the above direct estimates, estimated sampling variances and the selected auxiliary variables, compute the FH estimators and their estimated MSEs.

Functions `eblupFH()` and `mseFH()` from `sae` R package ([Molina and Marhuenda, 2009](#)) return the EBLUPs and their MSE estimates under the FH model. The calls to these functions are:

```
eblupFH(formula, vardir, method="REML", MAXITER=100, PRECISION=0.0001, data)
mseFH(formula, vardir, method="REML", MAXITER=100, PRECISION=0.0001, data)
```

Both functions require to specify the fixed part of the FH model (13) through a usual R `formula` object, placing the vector of direct estimates on the left-hand side of `formula` and the desired area level covariates separated by "+" on the right-hand side. As in any regression function in R, an intercept is added by default. These functions also require to specify the estimated sampling variances of the direct estimators in `vardir`. The default fitting method `method` is REML, but it can be changed to FH or ML. Default maximum number of iterations `MAXITER` and convergence tolerance criteria `PRECISION` of the Fisher-scoring algorithm can be also modified. The last argument, `data`, can be used optionally to specify a data object that contains the variables in `formula` and `vardir` as columns. The functions do not allow NA values because estimates are not returned for areas with zero sample size. Hence, these areas cannot be included in the data.

The function `eblupFH()` returns a list with two objects: `eblup`, a vector with the EBLUPs for the areas, and `fit`, which includes all interesting output from the fitting process. The function `mseFH()` gives also the EBLUPs, but supplemented with their estimated MSEs. This function returns a list with two objects: `est`, a list containing the EBLUPs and the results of the model fitting, and `mse`, a vector with the estimated MSEs.

Example 2 illustrates the calculation of FH estimators of area poverty rates at a given time t , using the functions `eblupFH()` and `mseFH()` of the `sae` R package (Molina and Marhuenda, 2009).

Example 2 (FH estimators of poverty rates, in R) *Continuing with the previous example, we illustrate how to obtain FH estimators of poverty rates for the provinces in R, following the FH estimation procedure included above.*

Step 1: Direct area estimators and their estimated sampling variances have been already computed in Example 1.

```
povinc.dir <- povinc.dir.res$Direct
```

First we analyze the normality assumption. While it should be verified using the model residuals, it would also be beneficial if the response variable itself was normally distributed. We will assess this by plotting a histogram of the direct estimates.

```
hist(povinc.dir,prob=TRUE,main="",xlab="Direct HT estimators pov. rate")
```

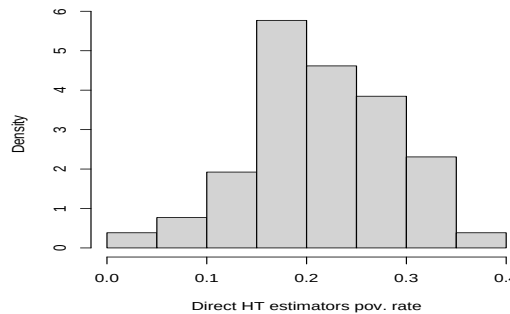


Figure 1: Histogram of direct estimates.

The shape of this histogram is somewhat skewed, but does not deviate significantly from a normal density, which is expected since the Central Limit Theorem applies to the direct estimators, as long as the area sample size n_d is not extremely small.

Optional part: Smoothing of direct estimators of variances.

```
povinc.dir.var<-povinc.dir.res$SD^2
lnd<-log(nd)
lvar<-log(povinc.dir.var)
fit.var<-lm(lvar~lnd)
sum<-summary(fit.var);sum
```

The result of this fit is:

Call:

```
lm(formula = lvar ~ lnd)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.13999	-0.25564	0.01994	0.29107	0.99673

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.09237	0.48896	-6.324	6.84e-08 ***
lnd	-0.70538	0.08822	-7.996	1.69e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5383 on 50 degrees of freedom

Multiple R-squared: 0.5611, Adjusted R-squared: 0.5524

F-statistic: 63.93 on 1 and 50 DF, p-value: 1.69e-10

We obtain smoothed variances as exponentials of the fitted values and plot the fitted values:

```
povinc.dir.svar<-exp(fitted(fit.var))
plot(nd,povinc.dir.var,xlab="nd",ylab="Estimated variance")
points(nd,povinc.dir.svar,col=2,pch=3)
```

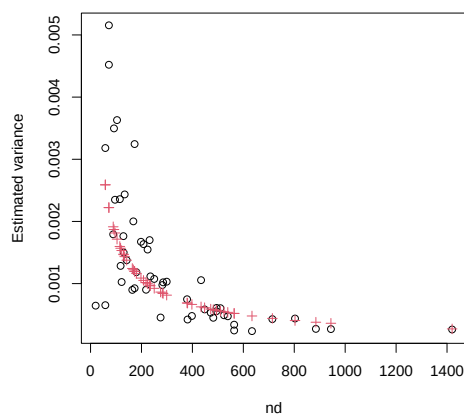


Figure 2: Raw sampling variances and smoothed variances against area sample size.

Step 2: We should apply a model selection procedure to find the area-level auxiliary variables that are useful for the FH model. Since in this example data are not real anyway, we just illustrate the calculation of FH estimates. Hence, we use all the available variables, excluding only those that would result in perfect multi-collinearity.

In a real application, it is advisable to perform a selection process of auxiliary variables, which is often done in the analogous linear regression model without the area effects (using the 'lm' function).

Next we load the data sets with the population sizes of the provinces. We also load the population sizes by nationality, age, and employment status, from which we will obtain the province proportions that will be used as auxiliary variables.

```
data(sizeprov)
data(sizeprovnat)
data(sizeprovage)
data(sizeprovedu)
data(sizeprovlab)
```

We now calculate the proportions of individuals in each category within each province, which will be the explanatory variables in a FH model:

```
Nd<-sizeprov[,3]
Ndnat<-as.matrix(sizeprovnat[, -c(1,2)])
Ndage<-as.matrix(sizeprovage[, -c(1,2)])
Ndedu<-as.matrix(sizeprovedu[, -c(1,2)])
Ndlab<-as.matrix(sizeprovlab[, -c(1,2)])

Pdnat<-Ndnat/Nd
Pdage<-Ndage/Nd
Pdedu<-Ndedu/Nd
Pdlab<-Ndlab/Nd

# Design matrix for the FH model
X<-cbind(const=rep(1,D), nat1=Pdnat[,1], Pdage[,3:5], Pdedu[,c(1,3)],
  Pdlab[,c(2,3)])
```

Step 3: We fit the FH model and verify the corresponding model assumptions. This includes assessing linearity, the normality of predicted area effects and standardized residuals, and identifying potential outlying areas. The function `eblupFH()` fits the model and already calculates the FH estimators of poverty rates by province. This function requires the direct HT estimators obtained in Example 1 and their estimated sampling variances. We consider now the smoothed sampling variances:

```
povinc.FH.res<-eblupFH(povinc.dir~X-1, vardir=povinc.dir.svar)
povinc.FH<-povinc.FH.res$eblup
```

To assess linearity, we plot residuals against predicted values (EBLUPs) in a scatterplot. The following R code may be used:

```
resids <- povinc.dir-povinc.FH
x11()
plot(povinc.FH,resids, pch=19,xlab="Predicted values (EBLUPs)",ylab="Residuals")
```

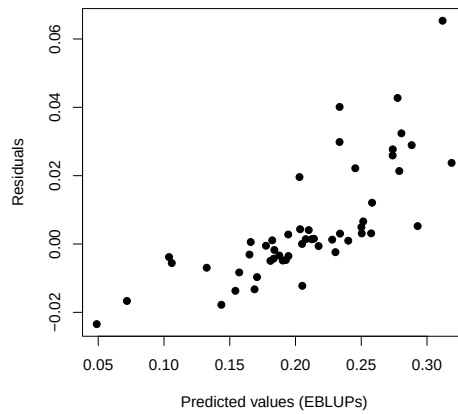


Figure 3: Residuals against predicted values (EBLUPs).

The plot seems to indicate some linearity issue in this case, which would require to change the model by including additional variables or changing the scales of those already included. A proper model selection procedure for the covariates might ameliorate or remove this non-linearity issue. In this example, we just intend to illustrate the application of the functions, so we continue with the same model. Normality of residuals and of predicted area effects may be checked visually with histograms or normal Q-Q plots, as follows.

```
x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(resids, probability=TRUE, main="",xlab="Residuals", breaks=6,
      ylim = c(0,23))
mean_est <- mean(resids)
sd_est <- sd(resids)
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(resids, main="")
```

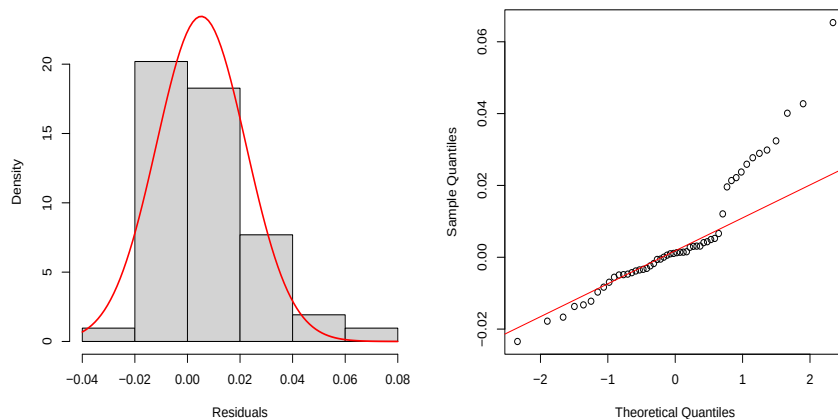


Figure 4: Histogram and normal Q-Q plot of model residuals.

Both plots indicate right skewness, showing a larger right tails than that of the normal density. The normality assumption might be also checked through a hypothesis test. In this example, we apply the Shapiro-Wilk normality test:

```
shapiro.test(resids)
```

Shapiro-Wilk normality test

data: resids

W = 0.88532, p-value = 0.0001221

This test indicates rejection of the normality assumption. As already said, normality is not actually required for the EBLUPs. Concerning the normality of the predicted area effects, since the function `eblupFH` does not give them in the output, here we first calculate them:

```
gammad <- povinc.FH.res$fit$refvar/(povinc.FH.res$fit$refvar+
                                   povinc.dir.res$SD^2)
random_effects <- gammad *(povinc.dir.res$X%*%povinc.FH.res$fit$estcoef$beta)
x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(random_effects, probability=TRUE, main="", xlab="Random Effects", breaks=6)
mean_est <- mean(random_effects)
sd_est <- sd(random_effects)
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(random_effects, main="")
shapiro.test(random_effects)
```

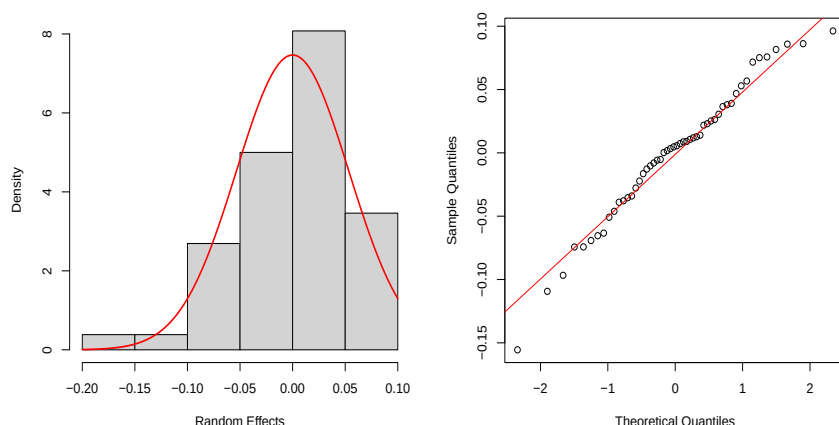


Figure 5: Histogram and normal Q-Q plot of predicted area effects.

Shapiro-Wilk normality test

data: random_effects

W = 0.97622, p-value = 0.38

The histogram and the Q-Q normal plot support the normality assumption for the predicted area effects and the Shapiro-Wilk does not reject normality.

Step 4: Since linearity actually fails, in a real application, the model should be changed.

Step 5: Once a satisfactory model is found, we calculate the FH estimates and the

associated estimated MSEs. In this case, we already have the FH estimates since the `eblupFH()` function fits the model and simultaneously provides the estimates.

For illustration only (not necessary in a real application), here we also calculate regression-synthetic estimators based using the estimated regression coefficients obtained from the model fitting:

```
povinc.rsyn1<-X%*%povinc.FH.res$fit$estcoef[,1]
```

Although these estimators are based on the regression coefficient estimates obtained from fitting the FH model rather than from a purely synthetic model, they are synthetic estimators since they do not account for unobserved heterogeneity between areas that is not explained by the auxiliary variables. Moreover, the regression coefficient estimates obtained under both models, using the same auxiliary variables, are asymptotically equivalent. Therefore, for large number of areas D , the two estimators will be very similar.

Since the FH estimators are a combination of the direct and the regression-synthetic estimators, it is interesting to observe the weights attached to the direct estimators. We have previously calculated these γ_d factors, a descriptive summary is given below:

```
summary(gammad)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4537	0.7182	0.8108	0.7906	0.8977	0.9477

We observe that, in this application, the weights assigned to the direct estimators are close to one for some provinces, implying small gains of EBLUPs with respect to the direct estimators.

Now we graphically compare the FH estimates with the direct HT estimates and synthetic estimates (labelled as RSYN1) for each province. The provinces (on the x -axis) are arranged in ascending order of sample size, and we indicate their sample sizes on the axis:

```
o<-order(nd)
k<-6
M<-max(povinc.dir,povinc.FH,povinc.rsyn1)
m<-min(povinc.dir,povinc.FH,povinc.rsyn1)
plot(1:D,povinc.dir[o],type="n",ylim=c(m,M+(M-m)/k),xlab="Province",
     ylab="Estimator",xaxt="n")
points(1:D,povinc.dir[o],type="b",col=1,lty=1,pch=1,lwd=2)
points(1:D,povinc.FH[o],type="b",col=4,lty=4,pch=4,lwd=2)
points(1:D,povinc.rsyn1[o],type="b",col=3,lty=3,pch=3,lwd=2)
axis(1, at=1:D, labels=nd[o])
legend(1,M+(M-m)/k,legend=c("DIR","FH","RSYN1"),ncol=3,col=c(1,4,3),
      lwd=rep(2,3),lty=c(1,4,3),pch=c(1,4,3))
```

Finally, we estimate the MSEs of the FH estimators by calling the function `mseFH()`. However, these MSE estimates are obtained under normality. Hence, we must exercise caution when looking at these MSE estimates. We calculate the CVs and plot the MSEs alongside the variances of the direct estimators:

```
povinc.dir.var <- povinc.dir.res$SD^2

povinc.FH.mse.res <- mseFH(povinc.dir~X-1,
vardir=povinc.dir.res$SD^2)

povinc.FH.mse <- povinc.FH.mse.res$mse
povinc.FH.cv <- 100*sqrt(povinc.FH.mse)/povinc.FH

M<-max(povinc.dir.var,povinc.FH.mse)
m<-min(povinc.dir.var,povinc.FH.mse)
```

```

plot(1:D,povinc.dir.cv[o],type="n",ylim=c(m,M+(M-m)/k),xlab="Province",
     ylab="CV",xaxt="n")
points(1:D,povinc.dir.var[o],type="b",col=1,lty=1,pch=1,lwd=2)
points(1:D,povinc.FH.mse[o],type="b",col=4,lty=4,pch=4,lwd=2)
axis(1, at=1:D, labels=nd[o])
legend(1,M+(M-m)/k,legend=c("DIR", "FH"),ncol=3,col=c(1,4),lwd=rep(2,2),
      lty=c(1,4),pch=c(1,4))

```

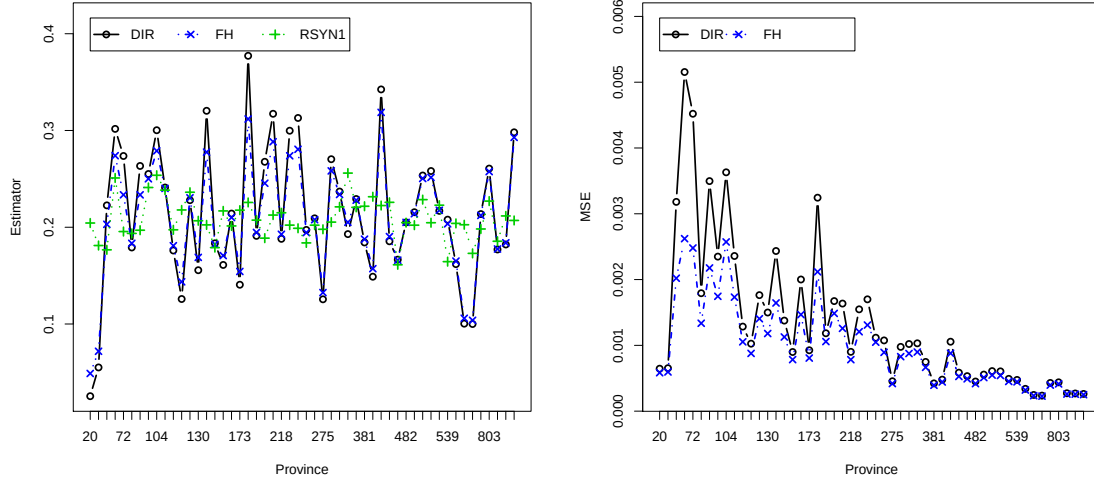


Figure 6: FH, direct HT and RSYN1 estimates of poverty rate for each province (left), and estimated MSEs of FH and direct HT estimators (right).

Once again, we can observe in Figure 5 (left) that the regression-synthetic estimators take very similar values across all provinces, in contrast with the direct estimators, which vary significantly across them. FH estimators approach the direct ones, but at the same time borrow information from other provinces through the synthetic estimators, especially for provinces with smaller sample sizes (left side of the plot). Even though the auxiliary variables in this example are very weak, Figure 5 (right) indicates efficiency gains for the FH estimators compared to the direct ones. Finally, we compare the estimated CVs for the direct HT and FH estimators, for the 5 provinces with the smallest sample sizes.

```

compardirFH<-data.frame(povinc.dir.cv, povinc.FH.cv)
selprov<-o[1:5]
compardirFH[selprov,]

```

The result is:

	povinc.dir.cv	povinc.FH.cv
42	99.97815	49.47213
5	46.35946	34.00564
40	25.33449	22.13933
34	23.80085	18.69105
44	24.57017	21.32242

We can see the reduction in CVs achieved by FH estimators compared to the direct HT estimators, with substantial gains for the two provinces with the smallest sample sizes.

5 Stable FH estimators over T time periods

This section describes procedures that yield stable small area estimators for each of D areas over T subsequent time instants. Since the area populations, the samples and the data might change between time periods, we now employ a more specific notation, adding subscript t to indicate elements for the t -th time instant ($t = 1, \dots, T$). Accordingly, we denote U_t the overall population at time t , which is partitioned into D areas $U_{1t}, \dots, U_{Dt}$, of respective population sizes $N_{1t}, \dots, N_{Dt}$. Now y_{dti} denotes the value of the target variable for the i -th unit at time t within area d , for $d = 1, \dots, D$, $t = 1, \dots, T$ and $i = 1, \dots, N_{dt}$. We wish to estimate an additive indicator (e.g. poverty rate) for each area d and time t , defined as

$$\delta_{dt} = \frac{1}{N_{dt}} \sum_{i=1}^{N_{dt}} h(y_{dti}), \quad d = 1, \dots, D, \quad t = 1, \dots, T. \quad (18)$$

To estimate those indicators, at each time t , a sample s_{dt} of size $n_{dt} < N_{dt}$ is drawn independently from each area d , yielding an overall sample $s_t = s_{1t} \cup \dots \cup s_{Dt}$, of size $n_t = \sum_{d=1}^D n_{dt} < N_t$, $t = 1, \dots, T$. We denote by $\pi_{dt,i}$ the inclusion probability of unit $i \in U_{dt}$ in the sample s_{dt} and $w_{dt,i} = \pi_{dt,i}^{-1}$ the corresponding survey weight. We further denote by $\pi_{dt,ij}$ the second-order inclusion probability of units $i \in U_{dt}$ and $j \in U_{dt}$ in s_{dt} .

Using one of the approaches described in Section 3, we can obtain direct estimators $\hat{\delta}_{dt}$ and their corresponding (estimated) sampling variances $\sigma_{ed,t}^2 = \widehat{\text{var}}_{\pi}(\hat{\delta}_{dt})$, for each area $d = 1, \dots, D$ at each time instant $t = 1, \dots, T$. However, for n_{dt} small, these estimators are very unstable across areas and over time, and might lead to very large/small estimates at time t and very small/large values at $t + 1$, which reduces their reliability.

FH estimates, $\hat{\delta}_{dt}^{FH}$, of δ_{dt} , and their estimated MSEs, $\widehat{\text{MSE}}(\hat{\delta}_{dt}^{FH})$, $d = 1, \dots, D$, may be obtained using a separate FH model for each time instant t , as described in Section 4, and then repeating the process for each $t = 1, \dots, T$. These estimates are stable across areas since FH model borrows strength across areas, but they need not be stable over time. Next we describe alternative multivariate FH models designed to provide more stable small area estimates over time.

As in the univariate FH model, the multivariate FH (MFH) model is defined in two stages. In the first stage, a linking model assumes that each target indicator δ_{dt} is linearly related with a vector $\mathbf{x}_{dt}$, containing the values of p_t auxiliary variables for area d and time t , that is,

$$\delta_{dt} = \mathbf{x}_{dt}' \boldsymbol{\beta}_t + u_{dt}, \quad d = 1, \dots, D, \quad t = 1, \dots, T. \quad (19)$$

In model (19), $\boldsymbol{\beta}_t$ is constant for all the areas but specific for each time instant, where u_{dt} is a specific random effect for area d and time instant t , with a distribution that will be detailed below. Let us now define

$$\boldsymbol{\delta}_d = \begin{pmatrix} \delta_{d1} \\ \vdots \\ \delta_{dT} \end{pmatrix}, \quad X_d = \begin{pmatrix} \mathbf{x}_{d1}' & 0 & \cdots & 0 \\ 0 & \mathbf{x}_{d2}' & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{x}_{dT}' \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\beta}_1 \\ \vdots \\ \boldsymbol{\beta}_T \end{pmatrix}, \quad \mathbf{u}_d = \begin{pmatrix} u_{d1} \\ \vdots \\ u_{dT} \end{pmatrix}.$$

Here, X_d is a $T \times p$ matrix, where $p = \sum_{t=1}^T p_t$ is the total number of auxiliary variables used for the T time instants. Using this notation, model (19) can be expressed as

$$\boldsymbol{\delta}_d = X_d \boldsymbol{\beta} + \mathbf{u}_d, \quad \mathbf{u}_d \stackrel{iid}{\sim} N(\mathbf{0}_T, \Sigma_u), \quad d = 1, \dots, D, \quad (20)$$

where $\Sigma_u = (\sigma_{u,ts})_{t,s=1}^T$ with diagonal elements also denoted by $\sigma_{u,t}^2 = \sigma_{u,tt}$, $t = 1, \dots, T$. Note that model (20) assumes a constant covariance matrix Σ_u for all the areas $d = 1, \dots, D$. In the second stage, $\hat{\delta}_{dt}$ is assumed to be unbiased for δ_{dt} , that is,

$$\hat{\delta}_{dt} = \delta_{dt} + e_{dt}, \quad d = 1, \dots, D, \quad t = 1, \dots, T. \quad (21)$$

Defining

$$\hat{\boldsymbol{\delta}}_d = \begin{pmatrix} \hat{\delta}_{d1} \\ \vdots \\ \hat{\delta}_{dT} \end{pmatrix}, \quad \mathbf{e}_d = \begin{pmatrix} e_{d1} \\ \vdots \\ e_{dT} \end{pmatrix},$$

we can express model (21) as

$$\hat{\boldsymbol{\delta}}_d = \boldsymbol{\delta}_d + \mathbf{e}_d, \quad \mathbf{e}_d \stackrel{iid}{\sim} N(\mathbf{0}_T, \Sigma_{ed}), \quad d = 1, \dots, D, \quad (22)$$

In model (22), $\Sigma_{ed} = (\sigma_{ed,ts})_{t,s=1}^T$, where $\sigma_{ed,ts} = \text{cov}_\pi(\hat{\delta}_{dt}, \hat{\delta}_{ds} | \boldsymbol{\delta}_d)$, $t \neq s$ and we also denote the variances by $\sigma_{ed,t}^2 = \sigma_{ed,tt}$, $t = 1, \dots, T$.

Replacing (20) in (22), we obtain a single model

$$\hat{\boldsymbol{\delta}}_d = X_d \boldsymbol{\beta} + \mathbf{u}_d + \mathbf{e}_d, \quad \mathbf{u}_d \stackrel{iid}{\sim} N(\mathbf{0}_T, \Sigma_u), \quad \mathbf{e}_d \stackrel{iid}{\sim} N(\mathbf{0}_T, \Sigma_{ed}), \quad d = 1, \dots, D. \quad (23)$$

Define additionally

$$\hat{\boldsymbol{\delta}} = \begin{pmatrix} \hat{\boldsymbol{\delta}}_1 \\ \vdots \\ \hat{\boldsymbol{\delta}}_D \end{pmatrix}, \quad \boldsymbol{\delta} = \begin{pmatrix} \boldsymbol{\delta}_1 \\ \vdots \\ \boldsymbol{\delta}_D \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ \vdots \\ X_D \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_D \end{pmatrix}, \quad \mathbf{e} = \begin{pmatrix} \mathbf{e}_1 \\ \vdots \\ \mathbf{e}_D \end{pmatrix}.$$

In matrix notation, the MFH model is then

$$\hat{\boldsymbol{\delta}} = X\boldsymbol{\beta} + \mathbf{u} + \mathbf{e}, \quad \mathbf{u} \sim N(\mathbf{0}_{DT}, V_u), \quad \mathbf{e} \sim N(\mathbf{0}_{DT}, V_e),$$

which is a special case of a linear mixed model, with block-diagonal covariance matrices (where each block is for an area), given by

$$V_u = \text{diag}_{1 \leq d \leq D}(\Sigma_u), \quad V_e = \text{diag}_{1 \leq d \leq D}(\Sigma_{ed}).$$

Note that, in the considered MFH model, the covariance matrix of the vector of response variables, $\hat{\boldsymbol{\delta}}$, is

$$V = \text{var}(\hat{\boldsymbol{\delta}}) = V_u + V_e = \text{diag}_{1 \leq d \leq D}(V_d), \quad V_d = \Sigma_u + \Sigma_{ed}, \quad d = 1, \dots, D.$$

For the special case of diagonal covariance matrices

$$\Sigma_{ed} = \text{var}(\mathbf{e}_d) = \text{diag}_{1 \leq t \leq T}(\sigma_{ed,t}^2), \quad \Sigma_u = \text{var}(\mathbf{u}_d) = \text{diag}_{1 \leq t \leq T}(\sigma_{ut}^2),$$

the MFH model (19)-(21) is the same as T separate (univariate) FH models, one for each time instant $t = 1, \dots, T$, which does not provide stable estimates over time. This means that applying separate (univariate) FH models for each time instant t actually assumes that area effects for the same area, $u_{d1}, \dots, u_{dT}$, and hence the corresponding target indicators $\delta_{d1}, \dots, \delta_{dT}$, are independent over time. This seems an unrealistic assumption, since the socio-economic conditions of a given area are not expected to strongly vary over time, unless strong interventions are made, in which case those interventions should be accounted for in the model (e.g. through adequate covariates). Here we consider two alternative models proposed by Benabent and Morales (2016), which incorporate correlation between the target indicators over time (time correlation), by considering full covariance matrices arising from AR(1) structure.

As in [Benabent and Morales \(2016\)](#), in the two alternative MFH models with time correlation, we consider general (full) covariance matrices for the model errors,

$$\Sigma_{ed} = (\sigma_{ed,ts})_{t,s=1}^T,$$

with known elements, for all $d = 1, \dots, D$. In practice, the covariances $\sigma_{ed,ts} = \text{cov}_\pi(\hat{\delta}_{dt}, \hat{\delta}_{ds})$, for each pair of time instants $t, s = 1, \dots, T$, are estimated from the actual survey sample data for area d , at time instants t and s , but they are regarded as the true covariances. Concerning the covariance matrix $\Sigma_u = \text{var}(\mathbf{u}_d)$ in the MFH model, we consider the alternative specifications proposed by [Benabent and Morales \(2016\)](#). They call Model 1 to the model obtained for Σ_{ed} full and Σ_u diagonal, that is,

$$\Sigma_u = \text{diag}_{1 \leq t \leq T}(\sigma_{ut}^2).$$

Model 2 is obtained with Σ_u resulting from an AR(1) process for $\mathbf{u}_d = (u_{d1}, \dots, u_{dT})'$,

$$\Sigma_u = \frac{\sigma_u^2}{1 - \rho^2} \begin{pmatrix} 1 & \rho & \dots & \rho^{T-1} \\ \rho & 1 & \dots & \rho^{T-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho^{T-1} & \rho^{T-2} & \dots & 1 \end{pmatrix}.$$

Finally, in Model 3, $\mathbf{u}_d = (u_{d1}, \dots, u_{dT})'$ follows the heteroscedastic AR(1) process

$$u_{dt} = \rho u_{d,t-1} + a_{dt}, \quad u_{d0} \sim N(0, \sigma_0^2), \quad a_{dt} \sim N(0, \sigma_t^2),$$

where $\sigma_0^2 = 1$, u_{d0} and $a_{d1}, \dots, a_{dT}$ are all independent, and where σ_t^2 , $t = 1, \dots, T$, are unknown time-specific model parameters. In contrast with Model 2, Model 3 leads to heteroscedastic time effects u_{dt} , $t = 1, \dots, T$, for each area $d = 1, \dots, D$.

If the samples s_{dt} drawn from area d at different time instants are independent, then it holds that $\Sigma_{ed} = \text{diag}_{1 \leq t \leq T}(\sigma_{ed,t}^2)$, $d = 1, \dots, D$. Often, rotating panels over time are employed in larger areas covering the small areas. If only a small portion of s_{dt} from the small area d is kept for the sample in the next period, $s_{d,t+1}$, we might decide to consider $\Sigma_{ed} = \text{diag}_{1 \leq t \leq T}(\sigma_{ed,t}^2)$ as an approximation of the true $\Sigma_{ed} = (\sigma_{ed,ts})_{t,s=1}^T$, $d = 1, \dots, D$.

The vector of BLUPs of $\boldsymbol{\delta} = (\boldsymbol{\delta}'_1, \dots, \boldsymbol{\delta}'_D)'$, for $\boldsymbol{\delta}_d = (\delta_{d1}, \dots, \delta_{dT})'$, is given for the three alternative MFH models by

$$\tilde{\boldsymbol{\delta}}^{MFH} = X\tilde{\boldsymbol{\beta}} + \tilde{\mathbf{u}},$$

where $\tilde{\boldsymbol{\beta}}$ is the BLUE of $\boldsymbol{\beta}$ and $\tilde{\mathbf{u}} = (\tilde{\mathbf{u}}_1, \dots, \tilde{\mathbf{u}}_D)'$, with $\tilde{\mathbf{u}}_d = (\tilde{u}_{d1}, \dots, \tilde{u}_{dT})'$ the vector of BLUPs of $\mathbf{u}_d$, given by

$$\tilde{\boldsymbol{\beta}} = (X'V^{-1}X)^{-1}X'V^{-1}\hat{\boldsymbol{\delta}}, \quad \tilde{\mathbf{u}} = V_uV^{-1}(\hat{\boldsymbol{\delta}} - X\tilde{\boldsymbol{\beta}}).$$

Note that the vector of BLUPs of $\boldsymbol{\delta}_d = (\delta_{d1}, \dots, \delta_{dT})'$ is given by

$$\tilde{\boldsymbol{\delta}}_d^{MFH} = X_d\tilde{\boldsymbol{\beta}} + \tilde{\mathbf{u}}_d, \quad d = 1, \dots, D,$$

where $\tilde{\mathbf{u}}_d$ is the vector of BLUPs of $\mathbf{u}_d = (u_{d1}, \dots, u_{dT})'$, given by

$$\tilde{\mathbf{u}}_d = \Sigma_u V_d^{-1}(\hat{\boldsymbol{\delta}}_d - X_d\tilde{\boldsymbol{\beta}}), \quad d = 1, \dots, D.$$

Note also that, by the block-diagonal structure of V , we can write

$$\tilde{\boldsymbol{\beta}} = \left(\sum_{d=1}^D X_d'V_d^{-1}X_d \right)^{-1} \sum_{d=1}^D X_d'V_d^{-1}\hat{\boldsymbol{\delta}}_d.$$

The vector of BLUPs $\tilde{\delta}^{MFH}$ of δ depends on a vector of unknown model parameters θ involved in the covariance matrix Σ_u , that is, $\Sigma_u = \Sigma_u(\theta)$ and $\tilde{\delta}^B = \tilde{\delta}^{MFH}(\theta)$, where θ varies across Models 1–3. Under Model 1, the unknown parameters are the T time-specific random effects variances σ_{ut}^2 , $t = 1, \dots, T$, that is, $\theta = (\sigma_{u1}^2, \dots, \sigma_{uT}^2)'$. Model 2 considers homoscedastic area-time effects over time, resulting in only two unknown parameters, σ_u^2 and ρ , that is, $\theta = (\sigma_u^2, \rho)'$. Finally, Model 3 considers heteroscedastic area-time effects over time, leading to $T + 1$ unknown parameters, given by σ_t^2 , $t = 1, \dots, T$, and ρ , that is, $\theta = (\sigma_1^2, \dots, \sigma_T^2, \rho)'$. The vector of EBLUPs $\hat{\delta}^{MFH}$ is defined by replacing θ by a consistent estimator $\hat{\theta}$ as $D \rightarrow \infty$ in the vector of BLUPs, that is, $\hat{\delta}^{MFH} = \tilde{\delta}^{MFH}(\hat{\theta})$. Benabent and Morales (2016) describe the REML fitting algorithm under each of these models, which leads to a consistent estimator $\hat{\theta}$ of θ in each model. Then, following the approach of Prasad and Rao (1990), Benabent and Morales (2016) also provide closed-form expressions for the mean squared errors of the EBLUPs $\hat{\delta}_{dt}^{MFH}$, $\text{MSE}(\hat{\delta}_{dt}^{MFH})$. Similarly, they provide estimators of those MSEs. They also consider bootstrap MSE estimators.

Below we list the steps for the calculation of MFH estimators for T time instants and their corresponding estimated MSEs.

MFH estimation procedure for T time instants:

- Step 1** Compute the selected direct area estimators $\hat{\delta}_{dt}$, $d = 1, \dots, D$, $t = 1, \dots, T$, and estimators of their corresponding sampling variances, denoted as $\sigma_{ed,t}^2$, and covariances, denoted $\sigma_{ed,ts}$, $d = 1, \dots, D$, $t, s = 1, \dots, T$.
- Step 2** Select the area-level auxiliary variables for each time instant in the MFH model. A simple approach is to perform a model selection procedure in a linear regression model without the area effects for each time instant $t = 1, \dots, T$.
- Step 3** Fit MFH3 model and test whether the area-time effects $(u_{d1}, \dots, u_{dT})$ are homoscedastic or not. If we reject the homoscedasticity of variances, consider MFH3 model. Otherwise, consider MFH2 model.
- Step 4** Check the selected model assumptions, including linearity, normality of predicted area effects and standardized model residuals, and the presence of outlying areas.
- Step 5** In case of clear systematic model departures, the model should be changed. In case of isolated departures because of outlying areas, either do not obtain the MFH estimate for those areas or change some aspect of the model or the data. Then, go to Step 2.
- Step 6** If model assumptions hold, using the above direct estimates and estimated sampling variances and covariances, and the selected auxiliary variables, compute MFH estimators $\hat{\delta}_{dt}^{MFH}$, $d = 1, \dots, D$ and $t = 1, \dots, T$, and their corresponding estimated MSEs.

The functions `eblupMFH2()` and `eblupMFH3()` from the R package `msae` (Permatasari and Ubaidillah, 2022) compute the EBLUPs and their MSE estimates under the MFH models 2 and 3, respectively. The calls to these functions are:

```
eblupMFH2(formula, vardir, MAXITER = 100, PRECISION = 1e-04, data)
eblupMFH3(formula, vardir, MAXITER = 100, PRECISION = 1e-04, data)
```

The arguments of these two functions are the same, differing only in the outputs. They require specifying a list with T R `formula` objects, one for each time instant, separated by commas. In each of these regression formulas (one for each time instant), we place the vector of direct estimates on the left-hand side and the area-level independent

variables on the right-hand side, separated by “+”. As usual, by default an intercept is automatically included in each regression formula.

As in the case of the `eblupFH()` function, they also require specifying the estimated sampling variances and covariances of the direct estimators for each area and time in the argument `vardir`. The user can also modify the maximum number of iterations, `MAXITER`, which is set by default to 100, and the convergence tolerance criteria, `PRECISION`, of the Fisher-scoring algorithm, which is set to $1e-4$. The final argument, `data`, can be optionally specified to indicate a data object that includes the variables found in `formula` and `vardir` as its columns. Similar to the `eblupFH()` function, these functions do not accept NA values, and they will not return estimates for areas with zero sample sizes. Consequently, such areas should be excluded from the dataset.

Both functions return a list containing the following objects: `eblup`, a vector of EBLUPs for the areas; `MSE`, a data frame with the estimated mean squared errors of the EBLUPs; `randomEffect`, a data frame containing the predicted area-time effects; `Rmatrix`, a diagonal matrix with the sampling errors; and `fit`, a list with additional information from the model fitting. In the `fit` list, we can find the fitting method used (`method`); a logical value indicating the convergence of the Fisher scoring algorithm (`convergence`); the number of iterations performed by the Fisher scoring algorithm (`iterations`); a data frame containing the estimated regression coefficients in the first column, their standard errors in the second, the t statistics in the third, and the p -values of the significance of each coefficient in the last column (`estcoef`); a data frame with the estimated random effects variance (`refvar`); a data frame with the estimated autocorrelation coefficient ρ of the random effects; and the estimated Fisher information matrix (`informationFisher`).

The function `eblupMFH3()` additionally includes in the `fit` list, a contrast to test the homogeneity of random effects variance, called `refvarTest`. This test helps the user to choose between the heteroscedastic Model 3 or the homoscedastic Model 2, for a particular data set.

Example 3 below illustrates the calculation of MFH estimators of area poverty rates for T time instants, using the functions `eblupMFH2()` and `eblupMFH3()` of the R package `msae` (Permatasari and Ubaidillah, 2022).

Example 3 (MFH estimators of poverty rates for T time periods, in R) *In this example, we use the data set `datasae3` from the R package `msae` (Permatasari and Ubaidillah, 2022). This data set contains simulated data generated under a FH model with heteroscedastic $AR(1)$ area-time effects. There are two auxiliary variables, `X1` and `X2`. Direct estimates for time instants 1, 2 and 3 are given in `Y1`, `Y2`, and `Y3`, respectively. The elements of the variance-covariance matrix of the sampling errors are given in `v1`, `v2`, `v3`, `v12`, `v13` and `v23`. We first load the package and the data set:*

```
library(msae)
data(datasae3)
```

Step 1: The direct area estimators and their sample variances are already given in the dataset, but the procedure to compute them was illustrated in Example 1.

Step 2: We should select the auxiliary variables at each time point. This example is only for illustration of application of the R function, and hence we use all the available variables, but we should emphasize the importance of this variable selection process in real-world applications.

Step 3: In order to choose between the two alternative MFH models with heteroscedastic or homoscedastic area-time effects, we fit the heteroscedastic Model 3, and then check for the equality of the variances over the three time points. Hence, we use the function `eblupMFH3` to fit the model:

```

Fo <- list(f1=Y1~X1+X2,
           f2=Y2~X1+X2,
           f3=Y3~X1+X2)
vardir <- c("v1", "v2", "v3", "v12", "v13", "v23")
m3 <- eblupMFH3(Fo, vardir, data=datasae3)

```

Next we check the equality of the random effect variances. For this, we use the results of a hypothesis test included in the output of the function `eblupMFH3`:

```

m3$fit$refvarTest
      refvar  T-test  p-value
1 Y1  vs  Y2 -0.88149  0.37805
2 Y1  vs  Y3 -2.8821  0.0039503
3 Y2  vs  Y3 -2.6816  0.0073271

```

This test tests the null hypothesis that the variances σ_t^2 at each pair of instants t and s are equal against the alternative that they are not. In this case, at significance level of 0.05, we reject the equality of variances between $t = 3$ and $t = 1$, as well as between $t = 3$ and $t = 2$. Note that this is a multiple test, and hence it is advisable to adjust the significance level. In this case, the tests clearly indicate heteroscedasticity, and we proceed with Model 3. If these tests supported equality of variances, then we would use instead the function `eblupMFH2` for estimation.

Step 4: We now verify the assumptions of the MFH3 model. This includes assessing linearity, the normality of the predicted area effects and standardized residuals, as well as checking for the presence of outlying areas. We first check the linearity assumption. This can be addressed by examining the scatter plot of residuals against predicted values (EBLUPs). The plot is generated for each of the $T = 3$ time periods.

```

resids_3 <- cbind(datasae3$Y1-m3$eblup$Y1, datasae3$Y2-m3$eblup$Y2,
                  datasae3$Y3-m3$eblup$Y3)
x11()
layout(matrix(1:3,nrow = 1, byrow = TRUE))
plot(m3$eblup$Y1,resids_3[,1],pch=19,xlab="EBLUPs t=1",ylab="Residuals t=1")
plot(m3$eblup$Y2,resids_3[,2],pch=19,xlab="EBLUPs t=2",ylab="Residuals t=2")
plot(m3$eblup$Y3,resids_3[,3],pch=19,xlab="EBLUPs t=3",ylab="Residuals t=3")

```

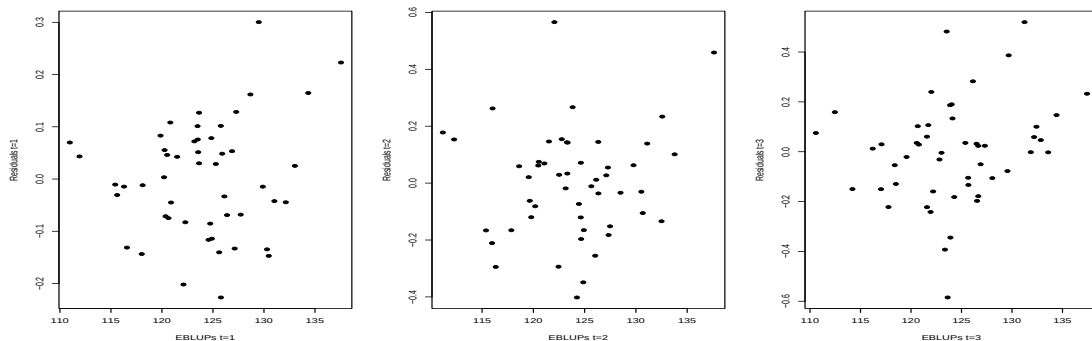


Figure 7: Residuals against predicted values (EBLUPs) for times $t = 1, 2, 3$.

These three plots do not provide evidences against the linearity assumption. We now evaluate the normality assumption of residuals, again for the 3 time periods:

```

x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(resids_3[,1], probability=TRUE, main="",xlab="Residuals t=1",
      ylim=c(0,3.7))
mean_est <- mean(resids_3[,1])
sd_est <- sd(resids_3[,1])
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(resids_3[,1], main="")
qqline(resids_3[,1], col="red")
shapiro.test(resids_3[,1])

```

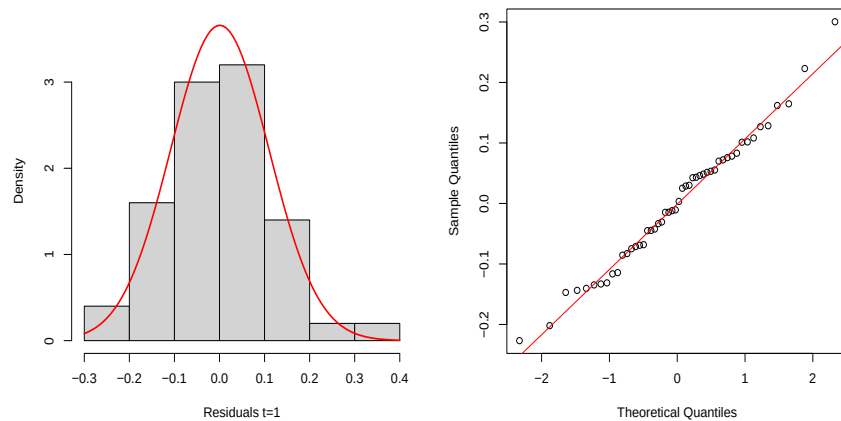


Figure 8: Histogram and normal Q-Q plot of the model residuals at $t = 1$.

```

Shapiro-Wilk normality test
data:  resids_3[, 1]
W = 0.98616, p-value = 0.8202

```

```

x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(resids_3[,2], probability=TRUE, main="",xlab="Residuals t=2")
mean_est <- mean(resids_3[,2])
sd_est <- sd(resids_3[,2])
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(resids_3[,2], main="")
qqline(resids_3[,2], col="red")
shapiro.test(resids_3[,2])

```

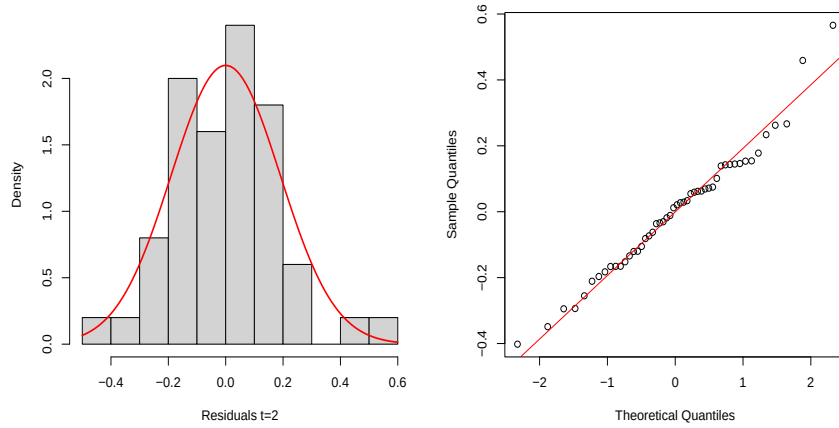


Figure 9: Histogram and normal Q-Q plot of the model residuals at $t = 2$.

Shapiro-Wilk normality test
data: resid_3[, 2]
W = 0.97649, p-value = 0.4151

```
x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(resids_3[,3], probability=TRUE, main="",xlab="Residuals t=3")
mean_est <- mean(resids_3[,3])
sd_est <- sd(resids_3[,3])
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(resids_3[,3], main="")
qqline(resids_3[,3], col="red")
shapiro.test(resids_3[,3])
```

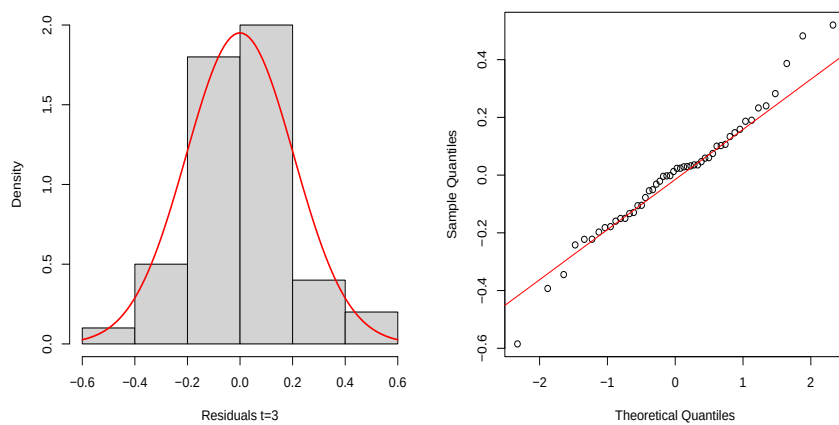


Figure 10: Histogram and normal Q-Q plot of the model residuals at $t = 3$.

Shapiro-Wilk normality test
data: resid_3[, 3]
W = 0.97556, p-value = 0.3833

In this case, the above histograms and Q-Q normal plots, as well as the Shapiro-Wilk tests, indicate nothing against the normality assumption of residuals. We next evaluate the normality of the random effects. Luckily, the function `eblupMFH3` provides the random effects in its output, for the three time periods:

```
ran_eff <- m3$randomEffect
x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(ran_eff$Y1, probability=TRUE, main="",xlab="Random Effects",
      ylim = c(0,0.45))
mean_est <- mean(ran_eff$Y1)
sd_est <- sd(ran_eff$Y1)
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(ran_eff$Y1, main="")
qqline(ran_eff$Y1, col="red")
shapiro.test(ran_eff$Y1)
```

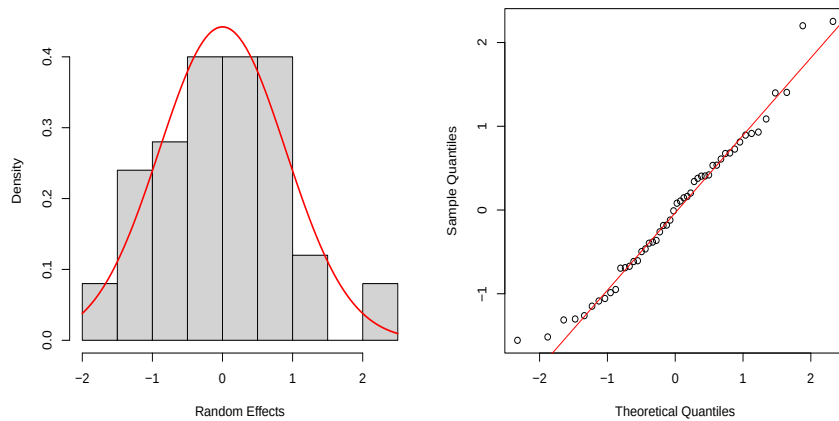


Figure 11: Histogram and normal Q-Q plot of predicted random effects at $t = 1$.

```
Shapiro-Wilk normality test
data:  ran_eff$Y1
W = 0.97539, p-value = 0.3778

x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(ran_eff$Y2,probability=TRUE,main="",xlab="Random Effects",breaks=5)
mean_est <- mean(ran_eff$Y2)
sd_est <- sd(ran_eff$Y2)
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(ran_eff$Y2, main="")
qqline(ran_eff$Y2, col="red")
shapiro.test(ran_eff$Y2)
```

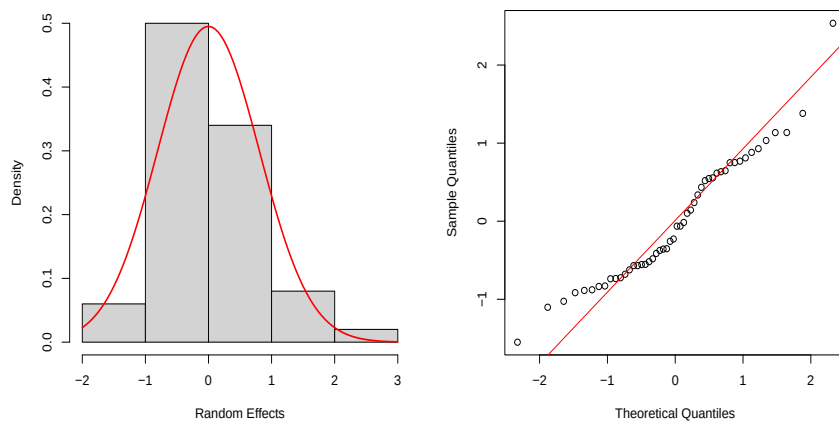


Figure 12: Histogram and normal Q-Q plot of the model random effects at $t = 2$.

Shapiro-Wilk normality test
data: ran_eff\$Y2
W = 0.95267, p-value = 0.04398

```
x11()
layout(matrix(1:2,nrow = 1, byrow = TRUE))
hist(ran_eff$Y3, probability=TRUE, main="",xlab="Random Effects")
mean_est <- mean(ran_eff$Y3)
sd_est <- sd(ran_eff$Y3)
curve(dnorm(x, mean=mean_est, sd=sd_est), add=TRUE, col="red", lwd=2)
qqnorm(ran_eff$Y3, main="")
qqline(ran_eff$Y3, col="red")
shapiro.test(ran_eff$Y3)
```

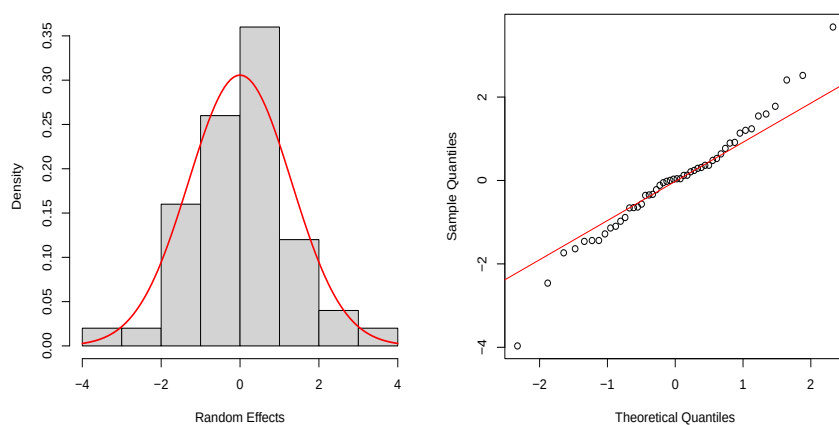


Figure 13: Histogram and normal Q-Q plot of the model random effects at $t = 3$.

Shapiro-Wilk normality test
data: ran_eff\$Y3
W = 0.97513, p-value = 0.3692

Again, histograms and Q-Q normal plots show no evidences of departure from normality, while the Shapiro-Wilk test also supports normality for all time periods at the usual significance level of 0.05, except for $t = 2$, which is supported at 0.01 level.

Step 5: Since no indications are found against the MFH3 model assumptions, we proceed to obtain the small area estimates, as well as their estimated MSEs. These are already been generated at output of the function together with the model fit, and can be printed as follows.

```
m2$eblup    # To see the EBLUPs
m2$MSE      # To see estimated MSEs of EBLUPs
```

We next illustrate the use of the function `eblupMFH2`, which should be used in the case of homoscedastic area-time effects over time. For this, we employ `datasae2` from the R package `msae` ([Permatasari and Ubaidillah, 2022](#)). In this new data set, again `X1` and `X2` are the auxiliary variables, and `Y1`, `Y2`, and `Y3` are the direct estimates at times 1, 2, and 3, respectively. The elements of the variance-covariance matrix of the sampling errors are provided in the variables `v1`, `v2`, `v3`, `v12`, `v13`, and `v23`. Again, we first load the package and the data set:

```
library(msae)
data(datasae2)
```

We now fit model MFH2 and check the model assumptions:

```
Fo <- list(f1=Y1~X1+X2,
           f2=Y2~X1+X2,
           f3=Y3~X1+X2)
vardir <- c("v1", "v2", "v3", "v12", "v13", "v23")
m2 <- eblupMFH2(Fo, vardir, data=datasae2)
m2$eblup    # To see the EBLUPs
m2$MSE      # To see the estimated MSEs of EBLUPs
```

The EBLUPs for each area and time period under the selected Model 3 are listed below:

```
m3$eblup
Y1      Y2      Y3
1  127.28 127.27 129.67
2  123.51 122.45 121.94
3  120.51 120.18 122.01
4  116.29 115.97 117.08
5  119.86 119.55 118.50
6  120.35 120.48 120.78
7  122.29 122.51 122.82
8  126.38 126.34 127.31
9  124.89 124.92 125.68
10 132.14 132.57 132.85
11 127.72 127.34 128.05
12 120.64 122.05 123.53
13 131.02 130.67 132.19
14 129.49 128.50 131.24
15 118.11 117.85 118.39
16 120.89 121.07 121.58
17 123.49 123.34 123.87
18 129.89 129.77 129.55
```

19	125.28	124.25	123.90
20	118.02	118.61	117.77
21	127.13	127.11	126.54
22	123.64	123.15	123.02
23	120.21	119.79	120.57
24	115.42	116.02	114.20
25	110.97	111.17	110.58
26	125.78	126.13	123.63
27	111.92	112.27	112.46
28	124.73	124.62	123.36
29	134.34	133.78	134.41
30	133.03	132.50	133.58
31	125.61	125.65	126.66
32	137.55	137.62	137.43
33	124.84	124.63	126.13
34	125.77	124.66	125.38
35	126.12	126.33	126.49
36	130.30	130.53	132.42
37	126.87	126.03	126.89
38	120.26	119.65	119.56
39	125.90	124.87	125.66
40	122.11	122.77	122.19
41	123.18	123.27	124.03
42	120.82	120.53	120.67
43	128.67	127.48	126.66
44	123.64	123.83	121.58
45	115.61	115.38	116.21
46	130.47	131.10	131.86
47	116.57	116.33	117.03
48	121.49	121.52	121.71
49	124.57	124.44	124.28
50	123.55	123.31	124.11

6 Spatio-temporal FH estimators over T time periods

This section describes an extension of the FH model by [Marhuenda et al. \(2013\)](#), which incorporates spatial correlation between neighboring areas and temporal correlation over T time instants, leading to more stable small area estimates over time.

Following the same notation as in the previous sections, δ_{dt} is the target area characteristic for area d , at time instant t , $\hat{\delta}_{dt}$ is the corresponding direct estimator and $\mathbf{x}_{dt}$ is a column vector with the values of p area-level auxiliary variables related linearly with δ_{dt} , for $d = 1, \dots, D$ and $t = 1, \dots, T$. The extended spatio-temporal FH (STFH) model is stated as follows. The sampling model is given by

$$\hat{\delta}_{dt} = \delta_{dt} + e_{dt}, \quad d = 1, \dots, D, \quad t = 1, \dots, T, \quad (24)$$

where, given δ_{dt} , sampling errors e_{dt} are assumed to be independent for all areas and time instants, and normally distributed with variances $\sigma_{ed,t}^2$, known for all d and t . Note that this model assumes that the covariances between direct estimators at different time instants is zero. The linking model is in this case

$$\delta_{dt} = \mathbf{x}_{dt}'\beta + u_{1d} + u_{2dt}, \quad d = 1, \dots, D, \quad t = 1, \dots, T, \quad (25)$$

where, as in the previous MFH Model 2, the vectors of area-time random effects $(u_{2d1}, \dots, u_{2dT})'$ are i.i.d. for each area d , following a (homoscedastic) AR(1) process with autocorrelation parameter ρ_2 , that is,

$$u_{2dt} = \rho_2 u_{2d,t-1} + \epsilon_{2dt}, \quad |\rho_2| < 1, \quad \epsilon_{2dt} \stackrel{iid}{\sim} N(0, \sigma_2^2). \quad (26)$$

Finally, the vector of area effects $(u_{11}, \dots, u_{1D})'$ follows a SAR(1) process with variance parameter σ_1^2 , spatial autocorrelation ρ_1 and row-standardized proximity matrix $\mathbf{W} = (w_{d,\ell})$, that is,

$$u_{1d} = \rho_1 \sum_{\ell \neq d} w_{d,\ell} u_{1\ell} + \epsilon_{1d}, \quad |\rho_1| < 1, \quad \epsilon_{1d} \stackrel{iid}{\sim} N(0, \sigma_1^2), \quad d = 1, \dots, D. \quad (27)$$

The proximity matrix $\mathbf{W}$ is obtained from an initial matrix $\mathbf{W}^0$, whose diagonal elements are equal to zero and the remaining entries are equal to 1 when the two areas corresponding to the row and the column indices are neighbors and zero otherwise. Then, row standardization is carried out by dividing each entry in $\mathbf{W}^0$ by the sum of the elements in the same row, see [Anselin \(1988\)](#), [Cressie \(1993\)](#) and [Banerjee et al. \(2004\)](#) for more details on the SAR(1) process with the above parametrization.

Consider the following vectors and matrices obtained by stacking the model elements in columns,

$$\begin{aligned} \hat{\delta} &= \text{col}_{1 \leq d \leq D} \left(\text{col}_{1 \leq t \leq T} (\hat{\delta}_{dt}) \right), \quad \mathbf{X} = \text{col}_{1 \leq d \leq D} \left(\text{col}_{1 \leq t \leq T} (\mathbf{x}'_{dt}) \right), \\ \mathbf{e} &= \text{col}_{1 \leq d \leq D} \left(\text{col}_{1 \leq t \leq T} (e_{dt}) \right), \quad \mathbf{u}_1 = \text{col}_{1 \leq d \leq D} (u_{1d}) \quad \text{and} \quad \mathbf{u}_2 = \text{col}_{1 \leq d \leq D} \left(\text{col}_{1 \leq t \leq T} (u_{2dt}) \right). \end{aligned}$$

Defining additionally the block-diagonal matrix $\mathbf{Z}_1 = \text{diag}_{1 \leq d \leq D}(\mathbf{1}_T)$, where $\mathbf{1}_T$ is a vector of ones of size T , and $\mathbf{Z}_2 = \mathbf{I}_m$, where $m = DT$ is the total number of observations, $\mathbf{u} = (\mathbf{u}'_1, \mathbf{u}'_2)'$ and $\mathbf{Z} = (\mathbf{Z}_1, \mathbf{Z}_2)$, the model can be expressed as a general linear mixed model of the form

$$\hat{\delta} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \mathbf{e}.$$

Observe that, here, $\mathbf{e} \sim N(\mathbf{0}_m, \mathbf{V}_e)$, where $\mathbf{0}_m$ denotes a vector of zeros of size m and $\mathbf{V}_e$ is a completely known diagonal matrix, given by $\mathbf{V}_e = \text{diag}_{1 \leq d \leq D}(\text{diag}_{1 \leq t \leq T}(\sigma_{ed,t}^2))$, since covariances between direct estimators at different time instants are not contemplated in this model.

We denote by $\boldsymbol{\theta} = (\sigma_1^2, \rho_1, \sigma_2^2, \rho_2)'$ the vector of unknown parameters involved in the covariance structure of the STFH model. Note that $\mathbf{u} \sim N(\mathbf{0}_m, \mathbf{V}_u(\boldsymbol{\theta}))$, where the covariance matrix is block-diagonal with two blocks, that is, $\mathbf{V}_u(\boldsymbol{\theta}) = \text{diag}\{\sigma_1^2 \Omega_1(\rho_1), \sigma_2^2 \Omega_2(\rho_2)\}$. Here, the matrices in the two blocks are respectively given by $\Omega_1(\rho_1) = \{(\mathbf{I}_D - \rho_1 \mathbf{W})'(\mathbf{I}_D - \rho_1 \mathbf{W})\}^{-1}$ and $\Omega_2(\rho_2) = \text{diag}_{1 \leq d \leq D}(\Omega_{2d}(\rho_2))$, with

$$\Omega_{2d}(\rho_2) = \frac{1}{1 - \rho_2^2} \begin{pmatrix} 1 & \rho_2 & \dots & \rho_2^{T-2} & \rho_2^{T-1} \\ \rho_2 & 1 & \ddots & & \rho_2^{T-2} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \rho_2^{T-2} & & \ddots & 1 & \rho_2 \\ \rho_2^{T-1} & \rho_2^{T-2} & \dots & \rho_2 & 1 \end{pmatrix}_{T \times T}, \quad d = 1, \dots, D.$$

Finally, the covariance matrix of the vector of response variables $\hat{\delta}$ is given by

$$\mathbf{V}(\boldsymbol{\theta}) = \text{var}(\hat{\delta}) = \mathbf{Z}\mathbf{V}_u(\boldsymbol{\theta})\mathbf{Z}' + \mathbf{V}_e.$$

Following [Henderson \(1975\)](#), the best linear unbiased estimator (BLUE) of β and the best linear unbiased predictor (BLUP) of $\mathbf{u}$ are given by

$$\tilde{\beta}(\theta) = \{\mathbf{X}'\mathbf{V}^{-1}(\theta)\mathbf{X}\}^{-1}\mathbf{X}'\mathbf{V}^{-1}(\theta)\hat{\delta} \text{ and } \tilde{\mathbf{u}}(\theta) = \mathbf{V}_u(\theta)\mathbf{Z}'\mathbf{V}^{-1}(\theta)\{\hat{\delta} - \mathbf{X}\tilde{\beta}(\theta)\}. \quad (28)$$

The second identity leads to the BLUPs of $\mathbf{u}_1$ and $\mathbf{u}_2$, which are respectively given by

$$\begin{aligned} \tilde{\mathbf{u}}_1(\theta) &= \sigma_1^2\Omega_1(\rho_1)\mathbf{Z}_1'\mathbf{V}^{-1}(\theta)\{\hat{\delta} - \mathbf{X}\tilde{\beta}(\theta)\} \text{ and} \\ \tilde{\mathbf{u}}_2(\theta) &= \sigma_2^2\Omega_2(\rho_2)\mathbf{V}^{-1}(\theta)\{\hat{\delta} - \mathbf{X}\tilde{\beta}(\theta)\}. \end{aligned}$$

Replacing an estimator $\hat{\theta}$ for θ in previous equations we obtain the so called empirical BLUE (EBLUE) $\hat{\beta} = \tilde{\beta}(\hat{\theta})$ and the empirical BLUPs (EBLUPs) of $\mathbf{u}_1$ and $\mathbf{u}_2$ respectively,

$$\hat{\mathbf{u}}_1 = \tilde{\mathbf{u}}_1(\hat{\theta}) = (\hat{u}_{11}, \dots, \hat{u}_{1D})' \text{ and } \hat{\mathbf{u}}_2 = \tilde{\mathbf{u}}_2(\hat{\theta}) = (\hat{u}_{211}, \dots, \hat{u}_{2DT})'.$$

Finally, the EBLUP of δ_{dt} under the STFH model (24)–(27), called more briefly STFH estimator, is given by

$$\hat{\delta}_{dt}^{STFH} = \mathbf{x}_{dt}'\hat{\beta} + \hat{u}_{1d} + \hat{u}_{2dt}, \quad d = 1, \dots, D, \quad t = 1, \dots, T.$$

The MSE of the STFH estimator may be estimated using the parametric bootstrap procedure proposed in [Marhuenda et al. \(2013\)](#).

Below we list the steps for the calculation of STFH estimators of δ_{dt} , for T time instants and their corresponding estimated MSEs.

STFH estimation procedure for T time instants:

- Step 1** Compute the selected direct area estimators $\hat{\delta}_{dt}$, $d = 1, \dots, D$, $t = 1, \dots, T$, and estimators of their corresponding sampling variances, denoted as $\sigma_{ed,t}^2$, $d = 1, \dots, D$, $t, s = 1, \dots, T$.
- Step 2** Select the area-level auxiliary variables for each time instant in the STFH model. A simple approach is to perform a model selection procedure in a linear regression model without the area effects for each time instant $t = 1, \dots, T$.
- Step 3** Fit the STFH model and check the model assumptions, including linearity, normality of predicted area effects and standardized model residuals, and the presence of outlying areas, for each time instant T .
- Step 4** In case of clear systematic model departures, the model should be changed. In case of isolated departures because of outlying areas, either do not obtain the STFH estimate for those areas or change some aspect of the model or the data. Then, go to Step 2.
- Step 5** If model assumptions hold, using the above direct estimates, their estimated sampling variances, and the selected auxiliary variables, compute STFH estimators $\hat{\delta}_{dt}^{STFH}$, $d = 1, \dots, D$ and $t = 1, \dots, T$, and their corresponding estimated MSEs.

EBLUPs for all areas and time instants, and parametric bootstrap MSE estimates can be obtained calling functions `eblupSTFH()` and `pbmseSTFH()` respectively. The calls to these functions are:

```
eblupSTFH(formula, D, T, vardir, proxmat, model = "ST", MAXITER = 100,
PRECISION = 0.0001, data)
pbmseSTFH(formula, D, T, vardir, proxmat, B = 100, model = "ST",
MAXITER = 100, PRECISION = 0.0001, data)
```


Some of the arguments are exactly the same as in the functions for FH model described in Section 4. Among the additional arguments, we have the number of areas D and the number of time periods T for each area. We remark that these functions may be used only when data are available for all the D domains at all T time periods. Moreover, data in `formula` and `vardir` must be sorted in ascending order by time instant, for each domain. Note that a single `formula` is specified in this function, unlike in the functions for the MFH modes of Section 5. The argument `model` can be chosen between the default value `ST` (AR(1) time-effects within each domain) or value `S` (with uncorrelated time effects within each domain). The two-standardized proximity matrix, $\mathbf{W}$, must be also given as input in `proxmat`. The elements of this matrix are in $[0,1]$, zeros on the diagonal and rows adding up to 1, as described above.

The function `pbmseSTFH()` providing bootstrap MSE estimates requires additionally to specify the number of bootstrap replicates B . A number of bootstrap replicates $B \geq 400$ is advisable to achieve stable MSE estimates. By default, the argument B is set to 100 to save computing time. To obtain the same MSE estimates every time the function `pbmseSTFH()` is run, the seed for random number generation should be fixed previously using `set.seed()`. The outputs of these functions have the same structure as the analogous functions for the FH model described in Section 4.

Example 4 illustrates the calculation of STFH estimators of area poverty rates for T time instants, using the above functions.

Example 4 (Spatio-temporal FH estimators of poverty rates, in R) *In this example, we use the data set `spacetime` included in the R package `sae`, which contains synthetic area level data for $T = 3$ time points, for each of $D = 11$ areas. The data set contains the following variables: `Area`, area code, `Time`, time point, `X1` and `X2`, the auxiliary variables for each area and time instant, `Y2`, direct estimates for each area and time instant, and `Var`, sampling variances of the direct estimators. We calculate EBLUPs of the means for each area at each time, based on the STFH model with proximity matrix given in the data set `spacetimeprox`. We also obtain the corresponding MSE estimates by parametric bootstrap. The steps of the STFH procedure described above should be followed but, in this example, we only illustrate the calculation of the small area estimators and their estimated MSEs. Calculation of direct estimates was illustrated in Example 1, and model diagnose was also illustrated in Examples 2 and 3.*

We first load the two data sets and obtain the number of areas and of time instants. Then, we apply the function `pbmseSTFH()` that delivers both, the STFH estimates and their estimated MSEs:

```
data("spacetime")
data("spacetimeprox")
D <- nrow(spacetimeprox)          # number of areas
T <- length(unique(spacetime$Time)) # number of time periods
set.seed(123)
STFH <- pbmseSTFH(Y ~ X1 + X2, D, T, vardir = Var, spacetimeprox,
                  data = spacetime)
```

The bootstrap procedure for MSE estimation displays the iteration number for each step, as follows:

```
Bootstrap procedure with B = 100 iterations starts.
b = 1
...
b = 100
```

Once we have obtained the STFH estimators, we compute their CVs, and the same is done for the direct estimators. We print the results for the last time instant ($T = 3$):

```

cv.STFH <- 100 * sqrt(STFH$mse) / STFH$est$eblup
cv.DIR <- 100 * sqrt(spacetime$Var) / spacetime$Y
results <- data.frame(Area = spacetime$Area, Time = spacetime$Time,
                      DIR = spacetime$Y, eblup.STFH = STFH$est$eblup,
                      cv.DIR, cv.STFH)
results.lasttime <- results[results$Time == 3, ]
print(results.lasttime, row.names = FALSE)

```

The obtained results for the last time instant are:

Area	Time	DIR	eblup.STFH	cv.DIR	cv.STFH
2	3	0.261484	0.27343181	10.944523	7.653997
3	3	0.175358	0.17722992	7.777336	7.026746
8	3	0.096230	0.09653879	6.059391	5.567674
12	3	0.122160	0.13740348	21.904205	14.798918
13	3	0.294176	0.29129477	8.812059	6.657347
16	3	0.412106	0.31887378	13.584403	9.224897
17	3	0.057924	0.06912566	25.195980	20.314774
25	3	0.209146	0.17377084	15.411972	12.225196
43	3	0.148671	0.14398844	15.788815	14.700855
45	3	0.234361	0.22810227	9.550663	8.303303
46	3	0.137869	0.14354272	8.853735	8.355827

Next we plot the STFH and the direct estimates together for comparison. Additionally, we plot their corresponding CVs. The following R code generates both plots:

```

x11()
layout(1)
results.lasttime <- results.lasttime[order(results.lasttime$cv.DIR), ]
plot(results.lasttime$DIR, type = "n", xlab = "area (time=3)",
     ylab = "Estimate", ylim = c(0.05, 0.45), cex.axis = 1.5, cex.lab = 1.5,
     xaxt = "n")
axis(1, 1:11, results.lasttime$Area, cex.axis = 1.5)
points(results.lasttime$DIR, type = "b", col = 1, lwd = 2, pch = 1)
points(results.lasttime$eblup.STFH, type = "b", col = 4, lwd = 2, pch = 4)
legend("top", legend = c("Direct", "EBLUP STFH"), ncol = 2, col = c(1, 4),
     lwd = rep(2, 2), pch = c(1, 4), cex = 1.3)

plot(results.lasttime$cv.DIR, type = "n", xlab = "area (time=3)",
     ylab="CV", cex.axis = 1.5, cex.lab = 1.5, xaxt = "n")
axis(1, 1:11, results.lasttime$Area, cex.axis = 1.5)
points(results.lasttime$cv.DIR, type = "b", col = 1, lwd = 2, pch = 1)
points(results.lasttime$cv.STFH, type = "b", col = 4, lwd = 2, pch = 4)
legend("top", legend = c("Direct", "EBLUP STFH"), ncol = 2, col=c(1,4),
     lwd = rep(2, 2), pch = c(1, 4), cex = 1.3)

```

Figure 13 left shows the STFH estimators, together with the direct estimates for each area at the last time point, with areas sorted by increasing CVs of direct estimators. Figure 13 right shows the corresponding CVs. In this example, we can see that, even with a very small number of areas ($D = 11$) and time instants ($T = 3$) to borrow strength from, the STFH estimates follow closely direct estimates, but are slightly more stable than them, and have smaller estimated CVs for all the areas.

We also apply basic univariate FH models for each time instant, that is, cross-sectionally, to analyze the differences with the results obtained from the STFH model.

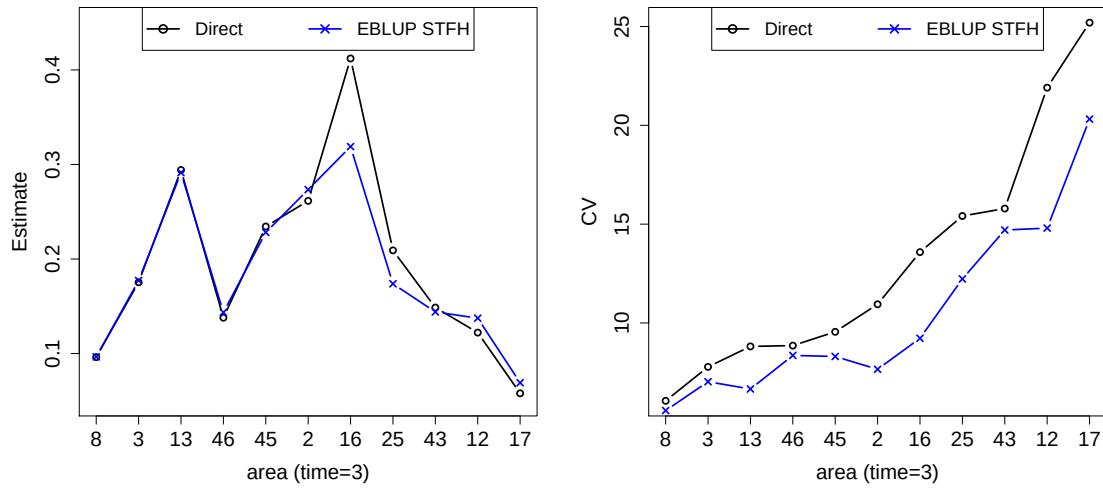


Figure 14: STFH and direct estimates of the area means at the last time instant $T = 3$ (left) and estimated CVs of the estimators for each area (right). Areas are sorted by increasing CVs of direct estimators.

For this, we first select the data corresponding to each time instant and then call the function `mseFH()` using those separate data sets. STFH estimates are stored in an additional column of the data frame `results`:

```
data.time1<-spacetime[spacetime$Time==1,]
data.time2<-spacetime[spacetime$Time==2,]
data.time3<-spacetime[spacetime$Time==3,]

eblup.FH.res.time1<-mseFH(Y~X1+X2,vardir=Var,data=data.time1)
eblup.FH.time1<-eblup.FH.res.time1$est$eblup
results$eblup.FH[results$Time==1]<-eblup.FH.time1

eblup.FH.res.time2<-mseFH(Y~X1+X2,vardir=Var,data=data.time2)
eblup.FH.time2<-eblup.FH.res.time2$est$eblup
results$eblup.FH[results$Time==2]<-eblup.FH.time2

eblup.FH.res.time3<-mseFH(Y~X1+X2,vardir=Var,data=data.time3)
eblup.FH.time3<-eblup.FH.res.time3$est$eblup
results$eblup.FH[results$Time==3]<-eblup.FH.time3
```

Let us now plot the STFH estimates, together with direct and cross-sectional FH estimates, for the first area (coded as `Area=2`) over the three time instants:

```
results.A1 <- results[results$Area == 2, ]

x11()
layout(1)
k<-3
m<-min(results.A1$DIR,results.A1$eblup.STFH,results.A1$eblup.FH)
M<-max(results.A1$DIR,results.A1$eblup.STFH,results.A1$eblup.FH)
plot(1:3,results.A1$DIR, type = "n", xlab = "Time",
     ylab = "Estimates: First area", ylim = c(m,M+(M-m)/k),
     cex.axis = 1.5, cex.lab = 1.5, xaxt = "n")
```

```

axis(1, 1:3, 1:3, cex.axis = 1.5)
points(1:3,results.A1$DIR, type = "b", col = 1, lwd = 2, pch = 1)
points(1:3,results.A1$eblup.FH, type = "b", col = 3,lwd = 2,pch = 3)
points(1:3,results.A1$eblup.STFH, type = "b", col = 4,lwd = 2,pch = 4)
legend("top", legend = c("Direct", "EBLUP FH", "EBLUP STFH"), ncol = 2,
      col = c(1,3,4),lwd = rep(2,3), pch = c(1,3,4), cex = 1.3)

```

Finally, we repeat the process for the last area (coded as Area=46):

```

results.A11 <- results[results$Area == 46, ]

x11()
layout(1)
k<-3
m<-min(results.A11$DIR,results.A11$eblup.STFH,results.A11$eblup.FH)
M<-max(results.A11$DIR,results.A11$eblup.STFH,results.A11$eblup.FH)
plot(1:3,results.A11$DIR, type = "n", xlab = "Time",
     ylab = "Estimates: Last area", ylim = c(m, M+(M-m)/k),
     cex.axis = 1.5, cex.lab = 1.5, xaxt = "n")
axis(1, 1:3, 1:3, cex.axis = 1.5)
points(1:3,results.A11$DIR, type = "b", col = 1, lwd = 2, pch = 1)
points(1:3,results.A11$eblup.FH, type = "b", col = 3,lwd = 2,pch = 3)
points(1:3,results.A11$eblup.STFH, type = "b", col = 4,lwd = 2,pch = 4)
legend("top", legend = c("Direct", "EBLUP FH", "EBLUP STFH"), ncol = 2,
      col = c(1,3,4), lwd = rep(2,3), pch = c(1,3,4), cex = 1.3)

```

Results are shown in Figure 14. We did already see in the previous examples that direct estimators are unstable across areas. We can see that they are also the most unstable over time. Cross-sectional FH estimators applied for each time instant $t = 1, 2, 3$, are more stable across areas, but not necessarily over time. Figure 14 shows that STFH estimators are smoother over time than both, direct and cross-sectional FH estimates.

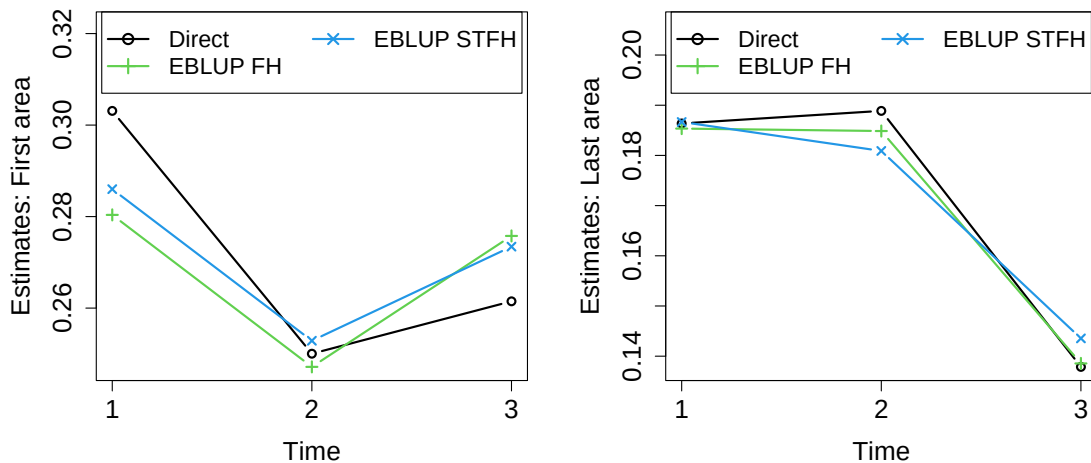


Figure 15: Direct, FH and STFH estimates of the area means for the first area (left) and the last area (right).

7 Statistical comparison of indicators over time

Now consider that, for a given area d , we wish to compare the indicators at two time periods, δ_{dt} and δ_{ds} , with $s < t$; more concretely, we wish to analyze the increment $\Delta_d = \delta_{dt} - \delta_{ds}$. Let $\hat{\delta}_{dt}^{EB}$ and $\hat{\delta}_{ds}^{EB}$ be EB predictors of δ_{dt} and δ_{ds} obtained either from the MFH of Section 5 or the STFH model of Section 6. Since there is dependency, making inference to compare δ_{dt} and δ_{ds} entails accounting for the uncertainty of the difference of EB predictors $\hat{\Delta}_d^{EB} = \hat{\delta}_{dt}^{EB} - \hat{\delta}_{ds}^{EB}$, which is given by

$$\text{MSE}(\hat{\Delta}_d^{EB}) = \text{MSE}(\hat{\delta}_{dt}^{EB}) + \text{MSE}(\hat{\delta}_{ds}^{EB}) - 2 \text{MCPE}(\hat{\delta}_{ds}^{EB}, \hat{\delta}_{dt}^{EB}).$$

Then, apart from estimators $\widehat{\text{MSE}}(\hat{\delta}_{dt}^{EB})$ and $\widehat{\text{MSE}}(\hat{\delta}_{ds}^{EB})$ of $\text{MSE}(\hat{\delta}_{dt}^{EB})$ and $\text{MSE}(\hat{\delta}_{ds}^{EB})$, an estimator $\widehat{\text{MCPE}}(\hat{\delta}_{dt}^{EB}, \hat{\delta}_{ds}^{EB})$ of $\text{MCPE}(\hat{\delta}_{dt}^{EB}, \hat{\delta}_{ds}^{EB})$ is required. Then, an estimator of the MSE of the difference is given by

$$\widehat{\text{MSE}}(\hat{\Delta}_d^{EB}) = \widehat{\text{MSE}}(\hat{\delta}_{dt}^{EB}) + \widehat{\text{MSE}}(\hat{\delta}_{ds}^{EB}) - 2 \widehat{\text{MCPE}}(\hat{\delta}_{ds}^{EB}, \hat{\delta}_{dt}^{EB}).$$

Although the distribution of the EB predictors is not really normal, a simple approximate $1 - \alpha$ confidence interval for the increment Δ_d , can be obtained using the normal quantile, as

$$\text{CI}_\alpha(\Delta_d) = \hat{\Delta}_d^{EB} \mp Z_{\alpha/2} \sqrt{\widehat{\text{MSE}}(\hat{\Delta}_d^{EB})}.$$

Using this interval, we can say that the difference Δ_d between indicators at times t and s is statistically different from zero at level α , if the $1 - \alpha$ confidence interval does not contain zero. Other hypotheses can also be tested about the increment Δ_d . Instead of considering the normal quantiles, an alternative approach is to use a bootstrap confidence interval.

The R package `msae` by [Permatasari and Ubaidillah \(2022\)](#) used in Section 5 and the functions for the STFH model of the R package `sae` by [Molina and Marhuenda \(2009\)](#) deliver the EBLUPs $\hat{\delta}_{dt}^{EB}$ of δ_{dt} , and their estimated MSEs, $\widehat{\text{MSE}}(\hat{\delta}_{dt}^{EB})$, $d = 1, \dots, D$, for each time point $t = 1, \dots, T$. Unluckily, these functions do not return estimates of the MCPE of two EB predictors. Again, a bootstrap procedure may be employed to estimate the MCPE and at the same time obtaining a confidence interval for Δ_d or performing a hypothesis test about Δ_d . Development and implementation of this bootstrap methodology would be needed.

8 Concluding remarks

This document describes implemented area-level SAE methods that provide statistically comparable and stable estimates over T time points, and estimators of their MSEs. The methods incorporate time correlation between the indicators at different time points, or spatial and temporal correlation. For each of the methods, the steps of the estimation procedures are listed, the available R functions that implement the methods are described and the application of the methods with those R functions is illustrated in examples that use free available data, so that users can reproduce all the results.

We note that models with time correlation need not improve in terms of efficiency to cross-sectional models applied at each time. Actually, more complex models require the estimation of additional parameters. In the case of area-level models, the more complex is the model, the larger number of areas is required to fit adequately the model. A small number of areas might lead to estimates of the model parameters out of their parameter space; e.g., negative variance estimates or autocorrelation parameters out of

the interval $(0,1)$. Hence, we recommend to apply the MFH or the STFH models when the number of areas is large.

Time stable empirical best (EB) predictors under a unit-level model with AR(1) time correlated area-time random effects were proposed by [Guadarrama et al. \(2021\)](#). The unit-level model is assumed for a transformation of a measure of welfare y_{dti} . Based on that model, EB predictors can be obtained for whatever additive indicator defined as in (18). Unluckily, user-friendly software for fitting the model and obtaining EB predictors, together with their MSE estimators, is not available. Hence, software development would be needed.

Making statistical inference (through confidence intervals or hypothesis testing) to compare indicators between two time points requires implementation of procedures for the estimation of mean crossed product errors (MCPE) of the two estimators. Even if this can be done e.g. using parametric bootstrap, it also requires software development.

Note also that making inference to compare indicators in two time points for one given area is fine, but temporal comparisons for a large number of areas encounters the problem of multiple testing. If we wish to test a hypothesis for each of the D areas, the significance level of each test should be corrected, using Bonferroni's correction, to α/D , in order to have a familiar level of α to reject at least one of the hypotheses. But since the number of areas D is supposed to be very large, α/D becomes too small, and then seldom rejecting any of the null hypotheses. Hence, testing hypotheses for many areas is not advisable.

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